



Review Article

A survey on quantum-safe blockchain security infrastructure

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ABSTRACT

Security infrastructure is vital in blockchain for its decentralized and distributed characteristics. Blockchain security infrastructure comprises several components: cryptographic algorithms, consensus protocols, key and identity management, network architecture, and smart contract deployment and execution. These components are vulnerable to the advancement of quantum computing and the realization of more powerful quantum computers. Classical security countermeasures used across the entire blockchain security infrastructure are exposed to quantum computing attacks. The exploitation is catastrophic to the sustainability of blockchain research and applications. This paper outlines the blockchain security infrastructure and quantum-safe solutions. The promising quantum-resistant cryptographic algorithms released by the National Institute of Standards and Technology (NIST) are evaluated for their relevance and use in blockchain security infrastructure. This paper also discusses the practical implementation and adoption of quantum-safe solutions for blockchain security infrastructure, including recent developments on the quantum-safe blockchain. The holy grail in adopting quantum-safe solutions for blockchain security infrastructure solutions is without sacrificing scalability and decentralization.

1. Introduction

Blockchain technologies allow decentralized and distributed applications to be developed without requiring central trusted intermediaries [1,2]. This is achievable through Distributed Ledger Technology (DLT) to store data and smart contracts to carry out functionalities automatically and autonomously [3,4]. Cryptographic algorithms, i.e., primitives and protocols: digital signatures and hash functions guarantee that all data stored as transactions in the ledger are immutable, valid, and non-repudiated [5,6]. As a decentralized and distributed system, the blockchain requires consensus protocols to achieve a universal synchronized view for all participants [7,8]. Hence, peer-to-peer (P2P) communication between participants must be carried out in a secured network environment, including the privacy of their identity through a secure key and identity management [9–11]. Due to the inexistence of central trusted intermediaries, functionalities for decentralized applications built atop blockchain technologies that use smart contracts rely on secure deployment and executions [12–14]. All these components are crucial for the operations and sustainability of blockchain.

The advancement of quantum computing and more powerful quantum computers pose existential threats to the future of classical cybersecurity [15–19]. Industrial standards cryptography algorithms, i.e., RSA (Rivest, Shamir, Adleman) and ECC (Elliptic Curve Cryptography),

are theoretically breakable by quantum computing [20,21]. These cryptographic algorithms, along with other cryptographic primitives in blockchain, are part of the security infrastructure that is helpless against quantum computing attacks [22]. Hence, it raises an open issue of blockchain security and the ability to embrace the quantum computing era. The race to create post-quantum cryptography (PQC) algorithms that could withstand quantum-based attacks was started by the NIST (National Institute of Standards and Technology) in 2016 [23]. Although one of its finalists, Rainbow, is cracked in 53 h using a standard laptop [24]. In July 2022, the federal agency revealed the first group of winners: CRYSTALS-Kyber, CRYSTALS-Dilithium, FALCON, and SPHINCS+ [25]. The NIST has recently started migrating to implement and adopt quantum-safe cryptography [26].

A systematic literature review on a quantum-safe blockchain system conducted in [27] shows the importance of creating a secure blockchain from quantum computing attacks. It reviewed 32 works and highlighted several works on digital signatures and public-key cryptosystems to thwart quantum computing attacks. Another review on blockchain asymmetric cryptography emphasized that the most prevalent public-key algorithms, e.g., RSA, ECDSA (elliptic curve digital signature algorithm), ECDH (elliptic curve Diffie-Hellman), and DSA (digital signature algorithm), are vulnerable to quantum computing attacks [21, 28]. A comparative analysis of the vulnerabilities of blockchain

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technologies to quantum computing in [21] analyzes major blockchain-based cryptocurrencies. The analysis concludes that existing major blockchain-based cryptocurrencies, e.g., Bitcoin and Ethereum, are vulnerable to exploitation by powerful quantum computers, i.e., Shor's quantum factoring algorithm [29] and Grover's quantum search algorithm [30].

Related studies [31–37] further signify the quantum-safe blockchain security infrastructure's importance in entering the quantum computing era. These studies suggest creating entirely new blockchain systems based on quantum-safe building blocks and approaches, i.e., quantum blockchain architecture, quantum digital signature and consensus, decentralized identity, quantum secure communication, and quantum blockchain development framework. Related work on securing blockchain networks in [38] pioneered a comprehensive framework for quantum-safe post-quantum blockchain networks that applies to extant blockchains. Each work contributed to particular topics of quantum-safe solutions directly or indirectly related to blockchain, i.e., quantum-safe key generation and distribution with quantum random number generators (QRNGs) [39–43] and quantum key distribution (QKD) protocols [44–49], quantum-resistant consensus mechanisms [50–55], quantum authentication protocols [56–59], and quantum internet protocols [60–63].

This paper intends to survey the current state-of-the-art works on blockchain security infrastructure, quantum threats and their impacts on the blockchain, and quantum-safe solutions. Existing review and survey articles on the quantum-safe blockchain are evaluated from the blockchain security infrastructure standpoint. Since most review and survey articles mentioned earlier focus only on specific parts of the quantum-safe blockchain systems. Thus, this paper presents the overview and state-of-the-art quantum-safe solutions for the blockchain security infrastructure, and the following are the contributions of this paper:

- It bridges blockchain technologies and quantum computing by performing an all-inclusive analysis of the overview of blockchain security infrastructure, quantum threats, and their impact.
- It compares existing studies on post-quantum cryptography (PQC) and various quantum-safe solutions related to blockchain security infrastructure to reveal state-of-the-art.
- It investigates the practical implementation and adoption of quantum-safe solutions for blockchain security infrastructure, which raises new research challenges.
- It provides future research directions to improve the quality of quantum-safe blockchains and focus on ongoing problems.

The remainder of this paper is structured as follows. [Section 2](#) outlines the survey's structure and methodology. [Section 3](#) compares related surveys with the survey presented in this paper. [Section 4](#) describes the overview of blockchain security infrastructure: cryptographic algorithms, consensus protocols, key and identity management, network architecture, and smart contract deployment and execution. [Section 5](#) lists quantum threats and their impact on the blockchain security infrastructure. [Section 6](#) reviews post-quantum cryptography (PQC) algorithms: lattice-based, code-based, multivariate, hash-based, isogeny-based, and fully homomorphic encryption (FHE) and their possible application in the blockchain. [Section 7](#) evaluates the practical implementation and adoption of quantum-safe solutions outside PQC for blockchain: quantum random number generators (QRNGs), quantum key distribution (QKD) protocols, quantum-resistant consensus mechanisms, quantum authentication protocols, and quantum internet protocols. [Section 8](#) explains the quantum safe blockchain alternatives. [Section 9](#) discusses the research challenges and trends in quantum-safe blockchain security infrastructure. [Section 10](#) concludes the paper by emphasizing the importance of quantum-safe blockchain security infrastructure.

2. Survey structure and methodology

The paper adheres to a particular sequence of steps to address the emerging domain of quantum-safe blockchain security infrastructure, as displayed in [Fig. 1](#). The survey begins with a comprehensive overview of the current state-of-the-art blockchain security mechanisms. This includes cryptographic algorithms, consensus protocols, key and identity management, network architecture, and smart contract deployment and execution, which form the foundation of secure blockchain operations. Next, the survey identifies quantum threats and assesses their impact on blockchain security, examining how advancements in quantum computing could compromise traditional cryptographic techniques and disrupt consensus mechanisms. The survey then explores post-quantum cryptography (PQC) and related quantum-safe solutions to address these threats, evaluating their potential for securing blockchain networks against quantum-enabled attacks. Following this, a review of related works on the practical implementation and adoption of quantum-safe solutions is conducted, analyzing existing research efforts and real-world applications that integrate quantum-resistant technologies into blockchain systems. Finally, the study discusses future research directions, identifying open challenges and opportunities for advancing quantum-safe blockchain security, ensuring long-term resilience against emerging quantum threats.

In order to enhance the organization of the survey, we adopt the taxonomy depicted in [Fig. 2](#). The taxonomy details the quantum-safe blockchain security infrastructure which is comprised of blockchain security infrastructure, quantum threats and their impact, post-quantum cryptography (PQC), quantum-safe solutions, and existing quantum-safe blockchains. The blockchain security infrastructure encompasses the fundamental security elements, including cryptographic algorithms, consensus protocols, key and identity management, network architecture, and smart contract deployment and execution. The quantum threats and their impact highlight the vulnerabilities posed by quantum computing, particularly its ability to break conventional cryptographic schemes. The post-quantum cryptography (PQC) explores cryptographic approaches designed to withstand quantum attacks, i.e., lattice-based, code-based, multivariate, hash-based, isogeny-based, and fully homomorphic encryption (FHE) approaches. The quantum-safe solutions extend beyond PQC to include technologies like quantum random number generators (QRNGs), quantum key distribution (QKD) protocols, quantum-resistant consensus mechanisms, quantum authentication protocols, quantum internet protocols. The existing quantum-safe blockchains highlights blockchain platforms that have integrated quantum-resistant technologies, serving as practical approach for analyzing the transition toward quantum-safe blockchain security, ensuring comprehensive coverage of threats, solutions, and ongoing developments in the field.

3. Related surveys

our survey papers related to quantum-safe blockchain security infrastructure have been identified and compared in [Table 1](#). All four works stress the importance of quantum-resistant, or post-quantum, cryptographic solutions as a means to secure blockchain infrastructures against future quantum attacks. They explore a range of emerging cryptographic approaches, including lattice-based and hash-based methods, which are seen as promising alternatives to current encryption techniques. The papers collectively highlight the urgency of rethinking blockchain security frameworks in anticipation of quantum computing advancements. Moreover, the papers delve into the broader implications of quantum threats on blockchain technology, particularly how these challenges affect consensus mechanisms, overall system performance, and structural integrity. They all point to the need for future research and development to bridge the gap between current blockchain practices and the requirements of a quantum-safe digital infrastructure.

This paper takes a broader, more infrastructure-centric approach

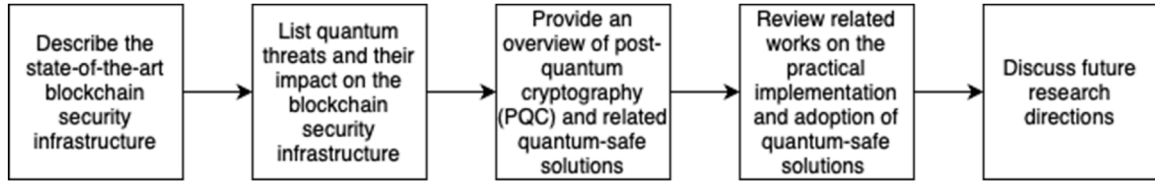


Fig. 1. The primary phases of establishing a quantum-resistant blockchain security framework and surpassing the existing state-of-the-art.

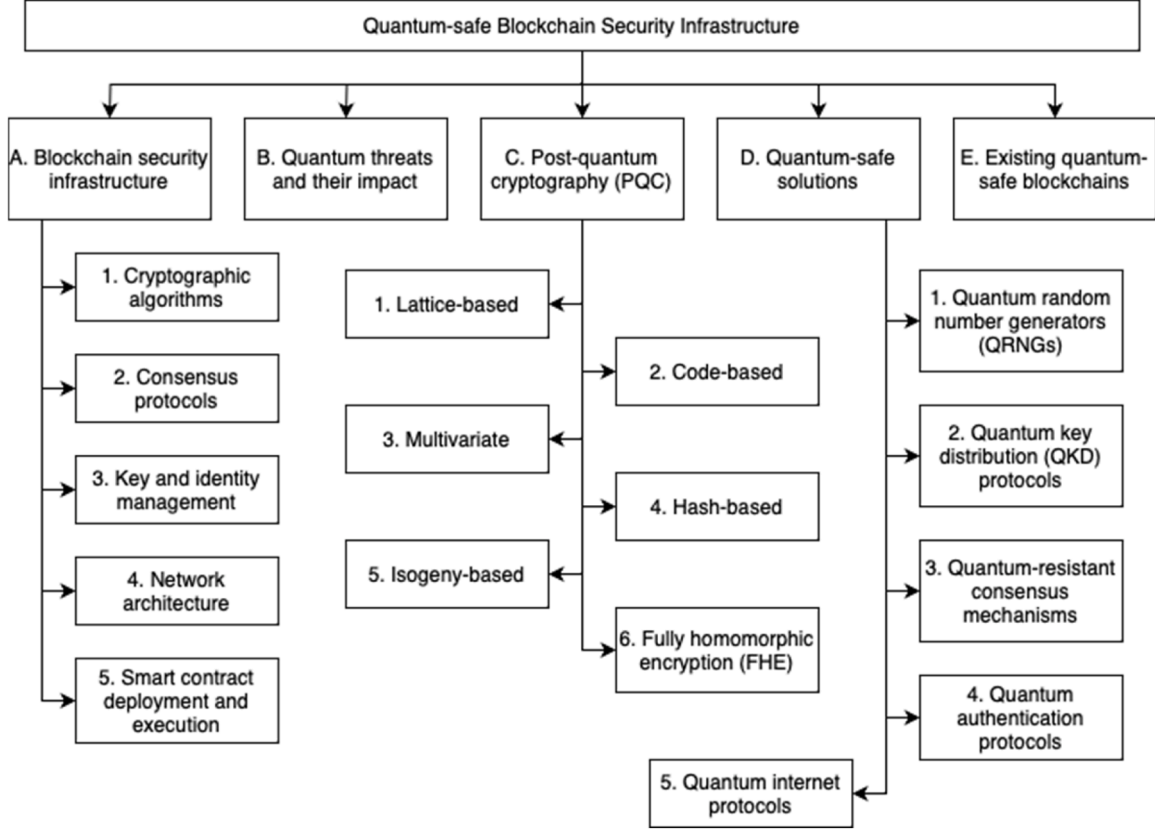


Fig. 2. Taxonomy of quantum-safe blockchain security infrastructure.

compared to the other four survey papers. While the four papers primarily focus on specific aspects of quantum threats to blockchain like trends in quantum-safe cryptography, tailored security frameworks for IoT environments, systematic reviews of post-quantum distributed ledger technologies, and direct comparisons between post-quantum and quantum blockchain approaches. This paper offers an all-inclusive analysis of blockchain security infrastructure. It covers not just cryptographic algorithms but also consensus protocols, key and identity management, network architecture, and smart contract deployment and execution, providing a comprehensive view of how each component is vulnerable to quantum attacks.

Another key difference lies in the methodology and scope. This paper surveys state-of-the-art quantum-safe solutions from 2019 to 2023 across various reputable sources, employing a taxonomy that categorizes quantum-safe approaches along the entire blockchain security stack. In contrast, the other papers tend to concentrate on narrower research questions—whether that’s evaluating emerging trends and vulnerabilities, proposing quantum blockchain frameworks for IoT, or directly comparing post-quantum and quantum blockchain paradigms.

Furthermore, while the four papers mainly address the quantum threat from the perspective of securing blockchain systems against cryptographic breakdowns (for example, by analysing the impact of Shor’s and Grover’s algorithms on RSA, ECC, and hash functions), this

paper expands the discussion to include the practical implementation and adoption challenges of quantum-safe solutions. It not only reviews the available post-quantum cryptographic algorithms (like those from NIST’s recent selections) but also discusses their relevance and potential trade-offs in terms of scalability and decentralization.

4. Blockchain security infrastructure

Security is viewed as one key component in blockchain, along with scalability and decentralization as described by the Blockchain Trilemma (Fig. 3), which is entangled and subject to a trade-off in implementation [68,69]. However, the trilemma focuses on scaling blockchain systems while maintaining decentralization and security [70]. Blockchain is a crypto-intensive system that depends fundamentally on cryptography building blocks, i.e., public-key cryptography and hash functions [71]. Public-key cryptography is used for digital signatures and encryption in the blockchain, with hash functions as the key technology for linking all the blocks together [5]. In addition, using zero-knowledge proof (ZKP) limits the amount of information transferred from prover P to verifier V [72], and it enhances privacy and anonymity, i.e., data confidentiality, transaction privacy, auditability without data exposure, selective disclosure, and confidential smart contract execution [73]. Additional cryptographic primitives, i.e.,

Table 1
Related surveys on quantum-safe blockchain security infrastructure.

Year	Refs.	Title	Outlook
2024	[64]	Quantum blockchain: Trends, technologies, and future directions	The authors provide a comprehensive survey of quantum blockchain developments. The paper outlines current trends, emerging technologies, and future directions in quantum blockchain. A central finding is that by integrating quantum-resistant cryptographic techniques (such as lattice-based methods) with classical blockchain architectures, one can both defend against quantum attacks and even harness quantum computing's capabilities for enhanced security and performance.
2024	[65]	Generic quantum blockchain-envisioned security framework for IoT environment: Architecture, security benefits and future research	The authors propose a generic security framework designed for IoT environments that leverages quantum blockchain technology. Their findings emphasize the architectural benefits and security enhancements possible when combining quantum cryptography (for example, quantum key distribution and quantum digital signatures) with blockchain principles. They also discuss potential applications and outline future research directions.
2024	[66]	A survey and comparison of post-quantum and quantum blockchains	The paper offers a detailed comparison between post-quantum and quantum blockchain approaches. It reviews the underlying cryptographic assumptions, blockchain structures, and consensus mechanisms. The findings highlight the strengths and weaknesses of each approach and discuss challenges in ensuring security and privacy during the transition to a quantum era.
2023	[67]	Post-quantum distributed ledger technology: a systematic survey	A systematic literature review is presented on post-quantum distributed ledger technologies (PQDLTs). This work examines the state-of-the-art cryptographic approaches that aim to secure blockchain systems against quantum attacks. The authors compare various post-quantum schemes and identify both performance challenges and open research questions in evolving PQDLT architectures.

special digital signatures, commitments, and accumulators, raise the privacy and anonymity of blockchain transactions [71]. Since communications occur between entities in the blockchain network, its security is crucial [74], including key and identity management [9–11]. Adding smart contracts into the blockchain requires secure deployment and execution to avoid malicious intent and exploitation [12,75]. The following sections outline blockchain infrastructure's key security elements: cryptographic algorithms, consensus protocols, key and identity management, network architecture, and smart contract deployment and execution.

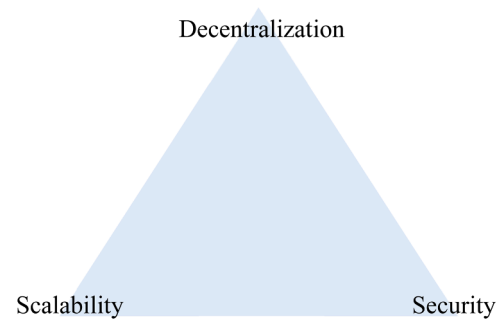


Fig. 3. The Blockchain Trilemma.

4.1. Cryptographic algorithms

4.1.1. Public-key cryptography

Blockchain technology generally employs various public-key cryptography algorithms for digital signatures and encryption [5]. Blockchain systems typically require efficient cryptographic algorithms due to the high volume of transactions and the need for fast verification. RSA encryption and decryption operations can be computationally intensive and slower than other encryption algorithms like ECC, especially with larger key sizes. Additionally, the key sizes required for RSA to provide sufficient security against modern computing capabilities can be larger, resulting in more significant storage and computational requirements [76], impacting the efficiency and scalability of blockchain networks, especially in decentralized and resource-constrained environments. RSA is based on the computational challenge of breaking down large integers into their constituent prime factors, which forms the basis of its security. Quantum computers could efficiently solve the factoring problem using Shor's algorithm, rendering RSA vulnerable to quantum attacks [15,77,78]. Instead of RSA, blockchain systems commonly employ ECC for their cryptographic operations [79].

Elliptic Curve Cryptography (ECC) offers comparable security to RSA but with lower key sizes and faster computational performance [80]. The security of elliptic curve cryptography (ECC) is based on the hardness of the elliptic curve discrete logarithm problem (ECDLP). The ECDLP forms the primary cryptographic assumption underlying the security of ECC [81]. The most commonly used ECC-based algorithms in blockchain to sign and verify transactions are ECDSA and EdDSA (Edwards-curve digital signature algorithm) [71,82]. ECDSA is used in Bitcoin, Ethereum, Litecoin, and Ripple for signing and verifying transactions [71]. EdDSA is used in Monero (Ed25519 variant) [21] and Cardano (Ristretto) [83] for anonymous signature verification and faster transaction processing. Moreover, they differ in their underlying fundamentals, performance characteristics, and security features presented in Table 2 [5,6,71,82,84].

Advanced cryptographic primitives in blockchains that could be deployed to enhance privacy and anonymity are special signatures (ring and multi-signature), commitments, accumulators, and zero-knowledge proofs (ZKPs) [71]. The notion of a ring signature enables the signing of messages on behalf of a group while maintaining the signer's anonymity. Key to the security of ring signatures is the computational impossibility of determining which specific group members used their keys to generate the signature. Unlike an individual signature, identity vulnerabilities cannot be exploited with a ring signature [85]. Due to the properties and mechanisms employed: ambiguity of the signer, computational infeasibility, unlinkability of keys, and unforgeability. Other variants of the ring signature are one-time and Borromean signatures [82].

The one-time ring signature combines the concept of one-time and ring signatures. Unlike the conventional one-time signature, this alternative variant exclusively permits linking two valid signatures if the same private key signs both. The Borromean ring signature (BRS)

Table 2
Comparison between ECDSA and EdDSA.

Criteria	ECDSA	EdDSA
Fundamental	Elliptic curves over prime fields generate digital signatures using point multiplication and modular arithmetic operations, forming the cryptographic foundation of ECDSA.	Twisted Edwards or Montgomery curves use algebraic operations that rely on the hardness of the elliptic curve discrete logarithm problem (ECDLP) and decisional Diffie-hellman (DDH), unlike prime-order curves.
Performance	Despite its computationally expensive modular arithmetic operations compared to EdDSA, ECDSA is widely supported and implemented in cryptographic libraries and standards due to its widespread implementation.	Enhanced performance compared to ECDSA due to reduced computations and faster signing and verification times, making it ideal for low-power devices and applications with limited computational resources.
Security	ECDSA is susceptible to side-channel attacks due to modular multiplication and point multiplication operations on elliptic curves and quantum attacks due to Shor's algorithm's ability to solve discrete logarithm issues.	EdDSA offers enhanced security and resistance against side-channel attacks, with shorter key sizes than ECDSA, simplifying key storage and distribution.

expands the capabilities of ring signatures to situations where the signer possesses knowledge of one secret from each group involved. The name "Borromean" is derived from the Borromean rings. In this mathematical concept, multiple interlocking rings are connected, so removing any individual ring causes the others to fall apart. This grants the ability to represent familiarity with any monotone Boolean function associated with the signing keys. Multi-signature (multi-sig) is a feature in blockchain technology that allows multiple parties to control the ownership and transfer of digital assets jointly. In a multi-sig scheme, a transaction requires the signatures of multiple participants to be considered valid and executed on the blockchain [71,86].

Commitments are cryptographic techniques that conceal the value or information being committed while allowing the commitment to be publicly verified later [71]. A commitment hides the original data but enables its verification against the committed value. This property helps preserve privacy and prevent data manipulation in various blockchain use cases. Commitments are often used in scenarios such as confidential transactions, where the value being transferred is hidden from public view while still allowing the verification of transaction integrity [87]. Commitments can also be used in decentralized applications (DApps) to demonstrate ownership of specific information without disclosing the specifics. Accumulators allow the aggregation of multiple data items into a single hash value, reducing the computational overhead of verifying individual elements [88]. In a blockchain context, accumulators are commonly used for maintaining the state of the blockchain, including account balances, transaction history, or smart contract states.

4.1.2. Zero-knowledge proof (ZKP)

Zero-knowledge proofs (ZKPs) are cryptographic protocols that allow a prover to demonstrate the veracity of a statement to a verifier without revealing any additional information. ZKPs possess three fundamental traits: completeness, soundness, and zero-knowledge. Completeness ensures an efficient means of verifying the accuracy of the statement. Soundness guarantees an extremely low probability of falsely confirming an incorrect statement [89]. Zero-knowledge proof ensures that, in a blockchain context, participants, excluding the data owner, remain oblivious to any supplementary details regarding private data [89]. The widely used ZKP protocols in the blockchain are zk-SNARKs (Zcash), zk-STARKs, and Bulletproofs (Monero) [71].

Zk-SNARKs achieve ZKP by generating succinct proof that the

verifier can efficiently verify. The prover creates a proof that satisfies certain properties, and the verifier can efficiently and quickly verify the proof without knowing any underlying details [89,90]. Unlike zk-SNARKs, which rely on a trusted setup and can be computationally intensive, zk-STARKs aim to achieve scalability and transparency by eliminating the need for a trusted setup and reducing computational requirements. Zk-STARKs achieve their properties through the use of transparent setup procedures and the application of concepts from interactive oracle proofs and error-correcting codes. They enable the prover to produce concise proof that the verifier can efficiently verify without requiring interaction between the two parties. One key advantage of zk-STARKs is their scalability [91].

Bulletproofs are specifically designed to prove the correctness of range proofs, demonstrating that a confidential value falls within a predetermined range without disclosing the precise value. Range proofs are commonly used in cryptographic systems like confidential transactions in blockchain networks to ensure that transaction amounts are valid and within specified bounds. One of the key features of bulletproofs is their logarithmic size. The magnitude of the proof grows logarithmically proportional to the number of proved constraints, resulting in significantly smaller proofs when compared to alternative forms of zero-knowledge proofs [92]. This logarithmic scaling enables efficient and scalable verification of proofs, reducing the computational overhead and improving overall performance. Bulletproofs are also non-interactive, which can be generated and verified without requiring interaction between the prover and verifier. This property allows flexibility and easier integration into various cryptographic protocols and systems [71]. Bulletproofs offer soundness and zero-knowledge properties, ensuring that the proofs are valid and provide no information about the underlying secret values. They achieve this through cryptographic commitments, inner-product arguments, and efficient inner-product proofs [71,92]. Table 3 compares widely used ZKP protocols in the blockchain [5,28,71,90–92].

4.1.3. Hash functions

A hash function is a mathematical operation that accepts input of any length ("message"). It produces a fixed-length output ("hash value" or

Table 3
Comparison of widely used ZKP protocols.

Criteria	zk-SNARKs	zk-STARKs	Bulletproofs
Fundamental	Succinct non-interactive proofs allow for verifying statements without revealing any details about them.	Transparent non-interactive proofs eliminate the need for a trusted setup and offer scalability for complex computations.	Efficient range proofs ensure that a value lies within a specified range without revealing the exact value.
Performance	Short proofs, resulting in efficient verification. However, the generation of proofs can be computationally expensive.	Longer proofs compared to zk-SNARKs, and the verification process can be more time-consuming.	Shorter proof sizes compared to zk-SNARKs and zk-STARKs, resulting in faster verification times.
Security	Rely on a trusted setup, which introduces a potential point of vulnerability. Zk-SNARKs are not quantum resistant due to the use of public-private key pairs.	Do not require a trusted setup, enhancing their security. Zk-STARKs are quantum resistant due to polynomial evaluation and polynomial degree bounds.	Non-interactive and do not rely on a trusted setup. They provide efficient range proofs and offer high security but primarily focus on a specific use case. Bulletproofs are considered to be not quantum resistant.

"digest"), which is typically a series of characters or numerals. The primary purpose of a hash function is to alter data in a predictable manner, which means that for the same input, the function will always generate the same output. Moreover, it should be computationally difficult to reconstruct the original input from the hash value result, making it a one-way function. Hash functions find wide application in various cryptographic scenarios, including blockchain systems, where they play a major role in ensuring data integrity and security. Notable characteristics of hash functions include determinism, fixed output size, resistance to preimage attacks (deriving the original input from the hash value), resistance to second preimage attacks (finding a different input yielding the same hash value), collision resistance (finding two distinct inputs that yield the same hash value), and the avalanche effect (a small change in the input should result in a substantial change in the hash value) [5]. The most popular hash function in the blockchain is SHA-256, and other widely used hash functions are Keccak-256, RIPEMD-160 (RACE Integrity Primitives Evaluation Message Digest), Blake2, and SHA-3 (Secure Hash Algorithm 3) [71].

Keccak-256 is an altered form of the Keccak hash function, which emerged as the victor of the NIST hash function competition in 2012 [93,94]. Ethereum, one of the most prominent blockchain platforms, initially used SHA-3 (another variant of Keccak) but later transitioned to Keccak-256 as the hash function for its addresses and hashing certain data structures [95]. RIPEMD-160 is often used in combination with other hash functions in blockchain applications. It is commonly used in Bitcoin and other cryptocurrencies for generating shorter address representations from public keys [96]. Blake2 is a cryptographic hash function that offers high performance and security [97]. Blake2 has several variants: Blake2b (64-bit platforms) and Blake2s (8- to 32-bit platforms), with different output sizes and security levels [98]. Its efficiency and speed have contributed to widespread adoption within blockchain systems, e.g., Monero and Siacoin [71]. Although less widely used and generally slower than SHA-256, SHA-3 (Keccak) is still utilized in some blockchain applications, e.g., Monero and Bytecoin [71]. It provides a different cryptographic construction compared to SHA-256 and offers resistance against specific types of attacks [99]. Table 4 provides an overview of the commonalities and distinctions of the aforementioned hash functions [5,71,94,97–100].

4.2. Consensus protocols

Cryptographic algorithms play a fundamental role in securing blockchain transactions by ensuring data integrity and authentication. However, as blockchain networks scale, the efficiency and resilience of these cryptographic mechanisms become increasingly critical, particularly in the face of emerging quantum threats. While cryptography

safeguards transaction validity, the broader security and functionality of blockchain systems also depend on effective consensus mechanisms. These protocols govern how distributed nodes reach agreement on transaction validity and block creation, ensuring the network remains secure, decentralized, and resistant to adversarial attacks. Given the interplay between cryptographic security and consensus mechanisms, understanding their relationship is crucial for designing resilient blockchain architectures in the post-quantum era.

A consensus protocol is a set of predetermined norms and procedures that facilitate agreement among a network's distributed nodes or participants on a single version of a shared state or data, even in the presence of faults, failures, or malicious behavior [101]. The consensus protocol serves several purposes in the blockchain: transaction validation, block creation and ordering, Sybil attack resistance, fault and Byzantine fault tolerance, decentralization, and security and immutability [102–104]. There is no best consensus mechanism but only the most suitable for a given application scenario [105]. Based on [5,8], the most widely used consensus protocols are proof of work (PoW), proof of stake (PoS), delegated proof of stake (DPoS), and practical Byzantine fault tolerance (PBFT). Other notable considerations are Hashgraph [106] and Directed Acyclic Graph [107] (DAG)-based ledger, which is built differently from the blockchain. Table 5 provides a comparison of the prevalent consensus protocols utilized in blockchain networks [5,8,101–104,108].

The PoW was invented by Adam Back in 1997 and developed initially as a Denial of Service (DoS) countermeasure against email spam [109]. Hal Finney, in 2004, adapted the PoW using 160-bit secure hash algorithm 1 (SHA-1) as reusable proof of work [110]. This consensus protocol was adopted by Satoshi Nakamoto in 2008 using SHA-256 for Bitcoin to achieve consensus in a permissionless and trustless environment [1]. The PoW governs block validation and addition into the chain through nonce value, transaction data, and hash function. Miners compete to find the nonce value as inputs with transaction data for a hash function with an output that matches the difficulty set by a number of leading zeros. The disadvantages of PoW are intensive computations and very low energy efficiency [1,5]. The PoW consensus protocol is designed to have high security and decentralization instead of scalability [105].

In contrast to PoW, which relies on computational labor and mining, PoS determines the next block's validator based on the stake or ownership of cryptocurrency held by network participants. Typically, the stake is determined by the number of currencies or tokens each player possesses. The more tokens a participant possesses, the greater the likelihood of being chosen as a validator [111]. The selected validator creates a new block containing a batch of transactions. The validator may include a cryptographic signature or proof of stake

Table 4
Overview of the most popular hash functions in blockchain.

Criteria	SHA-256	Keccak-256	RIPEMD-160	Blake2	SHA-3
Fundamental	Belongs to the SHA-2 family of hash functions, which utilizes the Merkle-Damgård construction to generate a consistent 256-bit fixed-size hash value.	Derived from the Keccak hash function, emerged victorious in the NIST hash function competition in 2012. It generates a 256-bit hash value based on the sponge construction principle.	Produces a 160-bit fixed-size hash value. It was built using the Merkle-Damgård construction principle.	Successor to the SHA-2 family of hash functions and offers several advantages in speed, security, and efficiency. Typically produces a fixed-size output of 256 bits.	Also known as Keccak, it is a family of cryptographic hash functions. SHA-3-256 produces a 256-bit hash value and is based on the sponge construction.
Performance	Good performance on most modern computing platforms.	High performance on various hardware platforms, including software and hardware implementations.	Generally considered to have moderate performance compared to other hash functions.	Designed to provide high performance on modern CPUs, including parallel and SIMD architectures. It is known for its fast speed and efficient implementation.	Good performance in general, comparable to SHA-256, on most modern computing platforms.
Security	Rely on the Merkle-Damgård construction, which assumes computationally impossible preimages, second preimages, or collisions for the compression function.	Utilize sponge construction for security, demonstrating resistance to preimage, second preimage, and collision attacks.	Rely on the Merkle-Damgård construction's resistance to withstand preimage and second preimage attacks.	Focus on finding preimages, second preimages, and collisions for its HAIFA construction.	Built on the Keccak sponge construction, ensuring resistance to preimage, second preimage, and collision attacks.

Table 5
Comparison of the most common consensus protocols.

Criteria	PoW	PoS	DPoS	PBFT
Fundamental	Blockchain protocol that uses miners to validate transactions by solving computational puzzles, requiring significant computational power and energy consumption.	A deterministic mechanism that selects validators to create new blocks based on their cryptocurrency holdings rather than computational puzzles.	The voting and delegation system allows token holders to elect delegates for block production and validation, combining PoS benefits with a delegated governance structure.	Consensus algorithm that can function even with malicious or failed nodes, requiring multiple rounds of message exchanges for consensus.
Performance	Resource-intensive and slower consensus mechanism due to its computational requirements for puzzle solving and longer block confirmation times.	Faster block confirmation times and higher transaction throughput compared to PoW, making it more energy-efficient.	Fast block confirmation times and high transaction throughput due to the smaller set of trusted delegates.	Fast consensus times and is known for its low latency, suitable for permissioned or private blockchain networks due to the assumption of a known set of participants.
Security	High-security level by requiring most of the network's computational power to attack the system. It has demonstrated robustness against various attack vectors, i.e., 51 % attack (majority attack), selfish mining, eclipse attack, Sybil attack, time jacking, and DoS.	Rely on stakeholders' economic incentives to maintain the network's security. Members with a larger stake are assumed to be more interested in maintaining the blockchain's integrity.	Rely on the assumption that elected delegates will act in the network's best interest. It introduces mechanisms for reputation systems and vote delegation to ensure accountability.	High-security level against Byzantine faults by requiring a two-thirds majority of honest nodes for consensus. It could withstand up to a third of malevolent or flawed nodes.

ownership. Other network validators confirm the validity of the newly created block. They verify the accuracy of transactions and ensure that the block's creator has the necessary stake to qualify as a genuine validator. As soon as a sufficient number of validators validate the block, it is added to the blockchain, and the procedure is repeated for the next block [112].

DPoS is an alternative to the traditional PoW and PoS consensus mechanisms. In DPoS, stakeholders can delegate their voting power to a trusted party or representative to participate in block production and validation on their behalf [113]. DPoS introduces a layer of delegation and decision-making by token holders. These elected validators are liable for generating new blocks and validating blockchain transactions. The number of validators is typically limited, making DPoS more scalable than PoW-based systems. DPoS block times (around 3s) are relatively short, allowing faster transaction processing than PoW-based blockchains with longer block times [111]. DPoS networks use a deterministic algorithm to resolve forks and reach consensus quickly. This feature reduces the possibility of chain splits and ensures a single, agreed-upon version of the blockchain [112].

PBFT is designed to achieve consensus in a network of nodes even when some nodes are faulty or malicious (Byzantine nodes). In a PBFT-based network, all nodes participate in the consensus process, and a consensus is reached through a series of rounds of communication and voting. The algorithm requires at least $3f + 1$ nodes to tolerate up to f faulty nodes, where f represents the utmost number of malfunctioning nodes the system can tolerate while maintaining consensus [8]. PBFT relies on a message-based protocol, where nodes exchange messages to propose and vote on blocks or transactions. Each node collects votes from the other nodes and waits for a sufficient number of affirmative votes to reach a decision. PBFT can achieve high throughput in contrast to consensus mechanisms like PoW and PoS. However, the overhead of message exchange increases with the number of nodes, which can limit its scalability in large networks [114].

4.3. Key and identity management

Consensus protocols establish trust and agreement among distributed nodes, ensuring the integrity and security of blockchain transactions. However, maintaining a secure and decentralized network requires more than just consensus; robust key and identity management systems are equally critical. These mechanisms control access to blockchain networks, protect private keys, and enable secure authentication for users and transactions. Without effective key and identity management, blockchain networks remain vulnerable to unauthorized access, identity fraud, and security breaches. As blockchain technology

continues to evolve, integrating strong cryptographic key management alongside consensus protocols is essential for ensuring both security and decentralization.

Key and identity management are two critical aspects of blockchain technology, particularly regarding securing and managing access to blockchain networks and resources. While they are related, they serve different purposes within the blockchain context. In the blockchain, key management refers to generating, exchanging, storing, using, rotating, and revocating cryptographic keys for authentication, encryption, and digital signatures. Keys are essential for secure communication, transaction signing, and access control within a blockchain network [9–11]. Recommendation for key management introduced by the NIST in [115] guides general key management and requirements for key protection. Blockchain key management systems are called wallets due to similar features to physical wallets in the traditional financial world.

The commonly used key management systems in blockchain could be categorized into three classes: paper wallets, hardware wallets, and software wallets. Paper wallets are the safest, storing the keys on paper as bar codes or quick response (QR) codes. The drawback of paper wallets is that the wallet is not immediately accessible (cold storage wallets). Hardware wallets are devices specifically designed to store and manage private keys securely. They offer offline storage and are considered one of the most secure methods for key management. Software wallets are applications or software programs installed on computers or mobile devices to manage private keys. They provide a convenient way to generate, store, and use cryptographic keys [116].

Key generation is a fundamental process in blockchain systems, responsible for creating the cryptographic key pairs that play a vital role in ensuring security and authenticity. During key generation, a private key is randomly generated using robust random number generators, safeguarding its secrecy as the owner's exclusive possession. A public key is derived from the private key using mathematical algorithms, establishing a unique relationship between the keys [117]. The public key is openly shared and often transformed into an address, providing a user-friendly transaction identifier in blockchain systems. In Bitcoin, the public key generates a wallet address for receiving funds. Similarly, in Ethereum, the public key generates an address for interacting with the Ethereum network, deploying smart contracts, and transferring Ether (the native cryptocurrency).

The key exchange in blockchain systems refers to securely sharing cryptographic keys between participants to establish secure communication channels and enable encrypted transactions. The Diffie-Hellman key exchange protocol, among others, is commonly used in blockchain, e.g., RSA and ECC. These protocols allow participants to generate a shared secret key without directly transmitting it over the network,

ensuring confidentiality and preventing eavesdropping or interception. In Bitcoin, the Diffie-Hellman and YAK are two key exchange protocols primarily used for transaction signing and verification [118]. When a user initiates a transaction, their private key is used to sign the transaction, and the corresponding public key is shared openly for verification. Similarly, in Ethereum, key exchange is crucial for interacting with the network, signing smart contracts, and validating transactions.

Key rotation involves periodically changing cryptographic keys used for various purposes. Key rotation helps mitigate the risk of key compromise and ensures the long-term security of the blockchain network. Regularly rotating keys can minimize the impact of a potential key compromise or vulnerability, as attackers would need to overcome the challenges of accessing and compromising the new keys [119]. Major blockchain systems like Bitcoin and Ethereum do not explicitly enforce key rotation, which introduces the risk of identity-stealing attacks once the corresponding private key is compromised. Key rotation ensures that stolen and leaked keys have a limited useful lifespan. Re-encrypted data using the new key are no longer accessible with the old key [120].

Key revocation requires invalidating and rendering previously issued cryptographic keys or certificates unusable. When a key is compromised, lost, or no longer trusted, key revocation ensures that the associated privileges and access rights are revoked, allowing dynamic access control. In blockchain networks, key revocation can be necessary when a participant's private key is compromised, when a user's access needs to be revoked, or when a key is suspected to be compromised due to a security breach. Key revocation methods and systems that existed are certificate revocation lists (CRLs), online certificate status protocols (OCSP), and Certificate Revocation Tree (CRT), which are commonly used to maintain and distribute the status of revoked keys across the network [121].

Identity management in blockchain involves the management of user identities and associated attributes within the network. It focuses on verifying and establishing participants' identities, ensuring their credentials' integrity, and managing access control to resources [122–124]. Bitcoin and Ethereum both lack an integrated identity management system. External services and entities may implement identity management systems for regulatory compliance, anti-money laundering (AML) measures, and know-your-customer (KYC) procedures [125,126]. Cryptocurrency exchanges and trading platforms frequently require users to provide personal information for identity verification.

Several identity management systems are self-sovereign identity (SSI), decentralized identifiers (DIDs), identity providers (IDPs), and OpenID Connect (OIDC) [127–129]. SSI is a concept that promotes individuals' ownership and control over their digital identities. Sovrin and uPort are examples of SSI projects that leverage blockchain for decentralized identity management. DIDs are unique, globally resolvable identifiers associated with entities (individuals, organizations, or devices) within a blockchain network. OpenID Connect is an OAuth 2.0 protocol-based identity layer. It allows users to authenticate with an identity provider and obtain an identity token to secure access to services and resources [128,130].

4.4. Network architecture

Effective key and identity management not only secure blockchain networks but also influence their overall architectural design. The structure of a blockchain network, including its level of decentralization and access control, is largely determined by its type, i.e., public, private, or consortium. Each type has distinct implications for how nodes communicate, how consensus is reached, and how security measures such as key management are implemented. Public blockchains prioritize decentralization and open participation, requiring extensive cryptographic safeguards, whereas private and consortium blockchains often implement more controlled identity management systems. Understanding these architectural differences is crucial for designing secure

and scalable blockchain ecosystems that align with the intended use case and security requirements.

4.4.1. Blockchain types

The type of blockchain significantly impacts the network architecture. The design and characteristics of different blockchain types influence how the network is structured, how consensus is reached, and how nodes communicate. Three recognized blockchain types are public, private, and consortium [131]. The network architecture of public blockchains requires many globally distributed nodes that communicate and replicate the entire blockchain's data, ensuring decentralization and consensus among participants. The network architecture of private blockchains may involve a smaller number of nodes that are known and authorized participants. Consortium networks can have different network structures depending on the requirements and agreements of the consortium members. Commonly, consortium networks are based on a peer-to-peer (P2P) architecture, where consortium members act as network nodes that communicate and share data. The differences between the three blockchain types are detailed in Table 6.

4.4.2. Blockchain nodes

Blockchain nodes are the individual participants or entities that comprise a blockchain network. Depending on the kind of blockchain and its consensus method, every node in a network of blockchains keeps a copy of the whole blockchain or just a subset of it. Nodes are essential to the blockchain network's functionality and security. The common types of nodes are full, light, mining, and validator nodes, as compared in Table 7 [132–134].

4.4.3. Network discovery

Network discovery is important in blockchain networks to foster decentralization, support peer-to-peer communication, facilitate consensus, enhance network scalability, and strengthen network security. It is vital in maintaining blockchain ecosystems' integrity, efficiency, and reliability. Network discovery in blockchain refers to finding and connecting to other nodes within a blockchain network. In network discovery, nodes employ various protocols and mechanisms to identify and connect with other nodes. These methods can include bootstrapping nodes, peer discovery protocols, DNS seed nodes, and peer exchange (PEX) [135,136].

In Ethereum, the network discovery process involves a combination of bootstrapping nodes and Ethereum Node Records (ENR). Ethereum relies on the Kademlia distributed hash table to manage its P2P network and for network discovery [137,138]. Bootstrapping nodes in Ethereum are pre-defined nodes that serve as initial connection points for new nodes joining the Ethereum network [139]. These bootstrapping nodes are hardcoded into Ethereum client software like Geth and Parity. Ethereum Node Records is a component of the Ethereum Wire Protocol and serves as a way for nodes to share information about themselves with other nodes in the network. ENRs contain essential information about a node, such as its IP address, port number, public key, network ID, and supported protocols. Nodes can exchange these records to discover and connect in the Ethereum network [140].

Bitcoin uses DNS seed discovery in addition to bootstrapping nodes and peer discovery protocols [141]. DNS seed nodes are pre-configured

Table 6
Blockchain types comparison.

Criteria	Public	Private	Consortium
Consensus	All users	Single user (owner)	Group of users
Permission	None	Yes (invitation)	Yes (invitation)
Scalability	Low	High	Medium
Security	Low	Medium	High
Privacy	High (anonymous)	Low (trusted)	Low (trusted)
Decentralization	Full	No	Partial

Table 7
Blockchain nodes comparison.

Criteria	Full	Lightweight	Mining	Validator
Fundamental	Ensure network integrity and decentralization by maintaining transaction history, verifying transactions, enforcing consensus rules, and relaying information.	Also known as thin or SPV nodes, they store a partial blockchain transaction history, prioritizing efficiency, and are suitable for devices with limited storage or computational capabilities.	Also known as miners, create new blockchain blocks by solving computationally intensive puzzles, adding them to the blockchain, and receiving rewards in cryptocurrencies like Bitcoin or Ethereum.	Crucial in PoS-based blockchain networks, participating in consensus processes by validating and proposing new blocks, using cryptocurrency holdings as collateral, and receiving rewards.
Performance	Store a comprehensive blockchain ledger, requiring significant computational resources and storage capacity for transaction validation and processing, making synchronization and processing time-consuming.	Use less computational resources and storage, reducing synchronization time but potentially introducing latency in accessing the latest blockchain data.	Requires significant computational power and energy to solve complex mathematical challenges, adding new nodes to the blockchain, which requires significant computational effort and time.	Play a crucial role in the blockchain network's consensus mechanism, requiring less computational power than mining nodes.
Security	Independently validating transactions and enforcing consensus rules, ensuring the integrity of the blockchain and contributing to its decentralization and security.	Relying on other nodes for data validation and accuracy, while full nodes trust them for accurate and validated information.	Participating in the Proof of Work consensus mechanism, validating transactions and blocks, and securing the blockchain against attacks like double-spending.	Participating in the consensus process, verifying transactions, and proposing new blocks.

domain names associated with the IP addresses of active Bitcoin nodes [142]. When a new node starts, it can query the DNS seed nodes to obtain a list of initial active nodes to connect. This initial connection helps bootstrap the node's connection to the Bitcoin network. In addition to DNS seed nodes, Bitcoin employs bootstrapping nodes, also known as seed nodes or hard-coded nodes [132]. These bootstrapping nodes are specific nodes whose addresses are hardcoded into the Bitcoin client software. New nodes connect to these bootstrapping nodes as their initial entry points to obtain information about other active nodes in the network and establish their initial connections [135].

4.4.4. Network communication protocols

Nodes communicate with each other using peer-to-peer (P2P) networking protocols, allowing them to propagate transactions and blocks, synchronize data, and reach consensus on the state of the blockchain [138]. In the Bitcoin network, nodes propagate transactions and blocks directly to their connected peers using a process known as flooding or broadcast. When a node receives a new transaction or block, it broadcasts that information to its connected peers. Each receiving node then verifies the received data and, if valid, further propagates it to its connected peers. This propagation continues throughout the network until the transaction or block reaches a significant portion of the nodes [141]. Other commonly used P2P communication protocols are the gossip and wire protocols [140].

The Gossip or flooding Protocol is a popular communication protocol in various blockchain networks. It allows nodes to disseminate information by randomly selecting and propagating data to a subset of their peers. The Gossip Protocol spreads information throughout the network, enabling efficient data distribution and consensus building [135]. Ethereum, for example, employs a variant of the Gossip Protocol known as the Ethereum Gossip Protocol for information dissemination among nodes. The Wire Protocol, also called the Network Protocol, is another widely used communication protocol in blockchain networks. It governs the low-level communication between nodes, including peer discovery, connection establishment, and message exchange. The Ethereum Wire Protocol (devp2p) is a prominent example, facilitating communication, block propagation, and network synchronization in the Ethereum network [140].

Protocols used within a blockchain network for purposes, i.e., facilitate secure and private computation on sensitive data across multiple nodes, affect P2P communication. Several popular protocols are Multi-Party Computation (MPC), ZKP, Byzantine fault tolerant (BFT), and inter-planetary file system (IPFS) [135,138]. MPC protocols allow nodes to jointly compute functions on sensitive data without revealing the actual data. In blockchain networks, MPC can be used for privacy-preserving operations, such as validating transactions or

performing computations on encrypted data [143]. BFT (Byzantine Fault Tolerant) Protocols: BFT protocols achieve consensus in distributed systems, including blockchain networks. Examples of BFT protocols include Practical Byzantine Fault Tolerance (PBFT) and Tendermint. These protocols allow nodes to agree on the order and validity of transactions, even when malicious actors or network failures are present [144]. IPFS (Inter-Planetary File System): IPFS is a peer-to-peer protocol and network designed to create a distributed file system. While primarily used for decentralized file storage, IPFS can also be leveraged for sharing data and content in blockchain networks [145,146].

4.4.5. External communication protocols

The blockchain network requires communication with external applications and services through standardized protocols and tools for querying blockchain data. The most commonly used methods and interfaces are remote procedure calls (RPCs), WebSockets, REST API, GraphQL, and message queuing protocols [147–156].

Remote Procedure Call (RPC): RPC is a communication protocol enabling nodes to interact, query data, and invoke remote procedures. It is commonly used for blockchain-related operations like submitting transactions and retrieving blockchain data. Bitcoin Core, the Bitcoin protocol's reference implementation, provides an RPC interface for developers to programmatically interact with the network [157].

WebSockets: WebSockets enables real-time bidirectional communication between clients and blockchain nodes. It provides a persistent connection, enabling efficient and low-latency communication for interactive applications. Ethereum nodes like Geth and Parity support WebSockets, enabling real-time communication with the Ethereum network. Developers can subscribe to events, receive notifications, and maintain a persistent connection for efficient data exchange [158].

REST API: Representational State Transfer (REST) application programming interface (API) enables standardized interactions between applications and services with blockchain networks, defining rules and protocols for external entities. These APIs can be used through HTTP/REST, WebSocket, or other communication protocols. Binance and Coinbase use RESTful APIs, providing seamless integration for trading functionalities, account information, order placement, and market data.

GraphQL: GraphQL is an indexing and querying protocol for blockchain data. It is an efficient alternative to REST for building APIs, allowing developers to query and manipulate data by requesting specific fields and relationships [159]. Avalanche, a high-performance blockchain platform, uses a GraphQL-based API to interact with the network, submit transactions, retrieve data, and access network functionalities [160].

Message Queuing Protocols: Message queuing protocols, e.g., Kafka, RabbitMQ, RocketMQ, ActiveMQ, Pulsar, AMQP (Advanced

Message Queuing Protocol), and MQTT (Message Queuing Telemetry Transport) enable asynchronous communication between external applications and blockchain networks, enabling scalable and loosely coupled patterns [156,161,162]. Hyperledger Fabric incorporates Kafka as an optional ordering service [163]. The permissioned R3 Corda blockchain uses AMQP and transport layer security (TLS) [164].

4.5. Smart contract deployment and execution

The architectural design of a blockchain network not only determines its level of decentralization and access control but also influences its capacity to support advanced functionalities such as smart contracts. As blockchain networks evolve, smart contracts have become a fundamental component, enabling automated execution of agreements without intermediaries. These self-executing contracts leverage the underlying blockchain infrastructure to ensure transparency, security, and immutability. However, the implementation and capabilities of smart contracts vary across different blockchain types, with script-based blockchains offering limited programmability and Turing-complete blockchains enabling more complex computational logic. Understanding these distinctions is essential for selecting the appropriate blockchain architecture for specific applications and use cases.

A smart contract is a contract that automatically executes, with the parameters of the agreement encoded in code. Once deployed on a blockchain, smart contracts execute predetermined actions autonomously based on the conditions specified in their code. Smart contracts automate the execution of agreements, eliminating the need for intermediaries and reducing manual processing time [13,14,165]. Since it operates on a decentralized and immutable blockchain, the code and the resulting transactions are visible to all participants in the network, creating a high level of transparency and eliminating the possibility of tampering with the contract terms [166]. The deployment of smart contracts could be categorized into (1) data storage contracts, (2) secure multi-party computation (MPC), (3) layer-2 protocols, (4) Scriptless contracts, (5) formal verification, (6) secure DevOps, and (7) containerization and sandboxing [13,14,167,168]. Existing blockchain platforms can be classified into script-based blockchains, exemplified by Bitcoin, and Turing-complete blockchains, exemplified by Ethereum. The former only supports limited expressions of operations (particularly no loops), whereas the latter supports arbitrary functions with Turing-complete programming languages [14,166,169]. Table 8 highlights the distinctions between script-based and Turing-complete blockchains in deploying smart contracts.

There are two major approaches to smart contract execution: on-chain and off-chain [170,171]. Most public blockchains store smart contracts on-chain that allow everyone to verify the content. All miners act following the transaction and contract and will ultimately reach a consensus on the execution results. The main disadvantage of this execution approach is privacy; for instance, trade secrets and confidential business processes [172]. Thus, this approach prevents executing certain commercial contracts [173]. The other approach is off-chain execution, which releases settling transactions only to the blockchain to improve performance and eliminate bottlenecks. This also conceals the contract's contents and user behavior details during data transmission, which offers better privacy than the on-chain execution approach [174]. Table 9 contrasts the execution of smart contracts between on-chain and off-chain approaches [14].

5. Quantum threats and their impact

Quantum computing is a rapidly advancing field of research that leverages the principles of quantum mechanics to perform complex calculations at an exponentially faster rate than classical computers [175–179]. Unlike classical computers, which use bits to represent information as 0 or 1, quantum computers use quantum bits or qubits, which can exist simultaneously in a superposition of both states [180].

Table 8

Comparison between script-based and Turing-complete blockchains.

Criteria	Script-based (Bitcoin)	Turing-complete (Ethereum)
Fundamental	Script-based blockchains typically use a scripting language that is limited in functionality and intentionally restrictive. In this context, a "script" refers to a simple set of instructions to validate and execute transactions on the blockchain. Bitcoin's scripting language is an example of a script-based blockchain.	Turing-complete blockchains, on the other hand, have a more expressive and versatile programming language, allowing developers to create complex and arbitrarily computable smart contracts. Ethereum is the most well-known example of a blockchain that is Turing-complete.
Complexity	The scripting language used in script-based blockchains is intentionally designed to be simple and restrictive. This simplicity ensures that scripts are predictable and safe, reducing the risk of unintended consequences and vulnerabilities.	The flexibility of the programming language in Turing-complete blockchains introduces higher complexity. Smart contracts can be more powerful and intricate, but this complexity can also increase the risk of vulnerabilities and exploits if not carefully developed and audited.
Security	Due to their limited functionality and restricted scripting language, script-based blockchains can be considered relatively secure and predictable. The simplicity of the scripts makes them easier to analyze, reducing the risk of unforeseen vulnerabilities.	Turing-complete blockchains require more careful attention to security best practices and auditing. The complexity of smart contracts increases the potential for bugs, vulnerabilities, and unintended consequences. Thorough testing, code review, and security audits are crucial to ensure the safety of Turing-complete smart contracts.

Table 9

Comparison between smart contract execution schemes.

Criteria	On-chain	Off-chain
Fundamental	The primary goal of private contracts with additional tools is to improve the capabilities and usefulness of permissioned or private blockchains. They function within a limited participant cohort and offer customization choices based on particular needs.	Off-chain channels let users share information and conduct direct transactions without depending entirely on the blockchain. They create transient private channels that facilitate scalable and effective transactions.
Complexity	The complexity of private contracts with extra tools can vary depending on the specific customization and additional tools implemented. They can introduce additional complexity due to custom consensus mechanisms or privacy protocols.	Off-chain channels complicate the transaction process, as participants must establish and manage temporary private channels. However, they typically offer simplicity in terms of the transactions conducted within the channels.
Security	The security of private contracts with additional tools depends on the design and implementation of the additional tools and the customization of the contracts. Attention must be given to ensure that the privacy features, scalability solutions, or custom consensus mechanisms do not introduce vulnerabilities.	The security of off-chain channels depends on the protocols and cryptographic mechanisms used to establish and secure the private channels. The risk lies in securing the off-chain transactions and final settlement on the blockchain.

Another key property of quantum computing is quantum entanglement. Entangled qubits are correlated with the state of another, regardless of the distance between them. These properties allow quantum computers to process vast amounts of information in parallel, enabling them to solve computationally infeasible problems for classical computers [181,

[182]. The following sections outline prominent quantum threats and their impact on the blockchain security infrastructure. Classical threats on blockchain systems that could be amplified by quantum computers, including new threats introduced by quantum computers, are presented.

5.1. Cryptographic algorithms

Shor's algorithm, developed by Peter Shor in 1994, is a quantum algorithm that can efficiently factor large numbers and solve the discrete logarithm problem [29]. This algorithm poses a significant threat to widely used cryptographic algorithms based on integer factorization, such as RSA and ECC [15,183,184]. Quantum computers with sufficient computational power can leverage Shor's algorithm to break these cryptographic schemes, compromising the security of sensitive information protected by these algorithms [15,78]. The computational complexity of the ECDLP is the basis for the security of ECDSA, which is extensively utilized in widely used blockchain systems, e.g., Bitcoin and Ethereum [38]. Quantum computers with Shor's algorithm have the potential to solve the ECDLP more efficiently than classical computers [28]. As a result, the security of ECDSA signatures could be compromised by a powerful enough quantum computer [185]. The break of ECDSA would compromise the security of transactions and other protected data. This could enable attackers to forge signatures, impersonate users, and tamper with transactions in blockchain networks. The alternative digital signature algorithm EdDSA used in blockchain systems like Monero is likewise susceptible to quantum computer attacks and risks similar impacts to the blockchain with ECDSA [21].

Grover's algorithm, proposed by Lov Grover in 1996, is a quantum algorithm that provides a quadratic speedup for searching unsorted databases [30]. Grover's algorithm could reduce the search space for collisions and preimage attacks, undermining the cryptographic properties of the hash functions [186–188]. Quantum computers can find collisions in hash functions more efficiently than classical computers, reducing time complexity to the square root of classical cases. This could break the resistance of hash functions against preimage attacks, which involve finding inputs that hash to a specific target hash value. Quantum amplitude amplification techniques can amplify the probability of finding a preimage, enhancing the chances of finding the desired preimage [189]. An algorithm developed in [190] targeted for hash preimage attacks has been proven to break preimage resistance with the idea of a time-space tradeoff to reduce the size of the whole circuit by running Grover's circuit multiple times. Attackers with sufficient quantum computers could forge transactions, manipulate block contents, or generate collisions, compromising transaction history and potentially causing blockchain systems disruption or loss of funds [21].

Grover's algorithm can also significantly speed up classical brute-force attacks by leveraging the parallelism and superposition properties of quantum computing. Quantum brute-force attacks could break ECC's security by exhaustively searching for the private key corresponding to a given public key, potentially leading to unauthorized access to encrypted data or the ability to create forged digital signatures. The quantum computer can search through many possible solutions in parallel and find the correct key or password with fewer iterations than classical methods. This reduces the time required to execute the brute-force attack, compromising the security of symmetric encryption schemes or password-based authentication systems [191]. It can also be applied to perform a brute-force search for preimage attacks, which involve finding an input that produces a specific hash value. Brute force attacks involve trying all possible inputs to find a match for a given hash value. If a quantum computer can search through many inputs in parallel, it can reduce the time required to find a match, making brute-force attacks more feasible. Grover's algorithm reduces the time complexity from exponential to the square root of the classical case, making it more efficient in finding preimages [17].

The use of zk-SNARK in blockchain applications relies on ECC as the foundation for the most effective zk-SNARK systems. These systems may

be used to demonstrate the accuracy of machine learning models and the accuracy of predictions because of their quick verification times and small proof sizes. However, for complicated models, the proving process requires an unreasonably high amount of memory and takes a long time. Moreover, none of these ECC-based systems are quantum resistant [192]. ZKPs' cryptographic primitives may become insecure, leading to the exposure of confidential information or transaction details. If verification relies on computationally hard problems, invalid proofs could be accepted, undermining the system's integrity [193]. The vulnerabilities posed by quantum computing extend beyond cryptographic primitives like zk-SNARKs to the very mechanisms that secure blockchain consensus, threatening the integrity and decentralization of the entire network.

5.2. Consensus protocols

Widely used consensus algorithms in blockchain are PoW, PoS, DPoS, and PBFT, which rely on cryptographic puzzles and algorithms [103,108,194]. Since these cryptographic assumptions are breakable by quantum computers, the integrity of the consensus process is compromised, leading to double-spending attacks or other consensus-related vulnerabilities [21,195,196]. Mining centralization and selfish mining are classical problems in PoW consensus protocol. Since quantum computers can solve mining puzzles significantly faster than classical computers, it would allow a single entity or a small group with quantum computing capabilities to control most of the network's mining power. This concentration of power undermines the decentralization and security of the blockchain network [197,198]. With a quantum advantage, attackers could mine blocks at a faster rate, potentially overtaking the legitimate chain and rewriting transaction history, leading to the possibility of double-spending attacks.

Quantum computers can amplify the 51 % attack. The increased computational power could enable an attacker to mine blocks and solve cryptographic puzzles much faster, allowing them to control most of the hash rate more quickly and execute a 51 % attack more efficiently. In a 51 % attack, an individual or group of attackers gain control over more than 50 % of the total computational power (hash rate) of the network [196,199]. In a Sybil attack, a malicious actor creates multiple identities or nodes to gain disproportionate control or influence over the network's operations, such as block production, voting, or decision-making processes. The attacker deceives the network by appearing as numerous independent participants when, in reality, they control a smaller number of entities [195,200,201]. Quantum computers do not directly amplify Sybil attacks against consensus algorithms since they involve an attacker creating multiple fake identities or nodes to gain control or influence over a network.

One of the security assumptions in PoS and DPoS consensus protocols is stake ownership, and the 51 % attack is not directly applicable. However, a long-range attack similar to the 51 % attack could rewrite the blockchain history by forging a blockchain from a past state using a significant stake in PoS and DPoS systems. This attack exploits the fact that stake ownership can change over time. There are three types of PoS long-range attacks: simple long-range, posterior corruption, and stake bleeding [195]. A problem in the PoS system is nothing-at-stake, which refers to the situation where validators can theoretically validate multiple competing chains without cost, leading to a lack of consensus and the potential for double-spending. This problem is addressed in DPoS systems by introducing the delegation model with the economic alignment of incentives as a deterrent against the nothing-at-stake problem.

PoS and DPoS systems are vulnerable to monopoly or stake concentration when a small number of participants or entities control a significant portion of the stake. This concentration of stake ownership could lead to collusion, manipulation, or unfair influence over the consensus protocol or decision-making process [112,202]. A particular problem with DPoS is the single point of failure due to the limited number of trusted delegates to validate transactions and secure the

network within the system. Thus, many unresponsive delegates would disrupt the consensus process, causing delays in block validation and leading to the single point of failure problem [111,194]. Beyond the risks to consensus mechanisms like PoS and DPoS, quantum threats further compromise blockchain security by targeting key management systems, potentially exposing private keys and undermining the cryptographic foundations that secure transactions and user identities.

5.3. Key and identity management

Quantum threats pose significant risks to blockchain key management systems, which rely heavily on cryptographic algorithms for generating, exchanging, and securing cryptographic keys. Quantum computers have the potential to break public-key cryptography algorithms like RSA and ECC, allowing malicious actors to forge signatures and impersonate legitimate users. Quantum threats also affect random number generators (RNGs) for key generation, as quantum computers can predict or influence the output of classical RNGs. Traditional RNGs rely on deterministic algorithms or physical processes, which could be vulnerable to attacks from quantum computers. A powerful quantum computer equipped with Grover's algorithm can search through possible seed values and break the security of these RNGs with a quadratic speedup, allowing malicious actors to predict or control the random numbers generated, compromise cryptographic keys used for encryption, digital signatures, and other security mechanisms in blockchain and other cryptographic systems [42,203,204].

Quantum threats also extend to the key exchange protocols used in blockchain systems, such as Diffie-Hellman and ECDH. Since these protocols rely on mathematical problems that quantum computers can efficiently solve, quantum computers can employ Shor's algorithm to efficiently factor large numbers, breaking the security of RSA-based key exchange. Similarly, quantum computers can rapidly solve the discrete logarithm problem, undermining the security of key exchange protocols like Diffie-Hellman and ECDH [17,205,206]. Thus, potentially compromising the confidentiality and secrecy of keys exchanged between parties. Consequently, encrypted communications within the blockchain network could be susceptible to interception and decryption by attackers with access to powerful quantum computers.

There are no known quantum threats against paper wallets. Unless an adversary obtains a paper wallet's private key, the associated funds could be accessed and potentially stolen. This poses a significant risk to individuals and organizations relying on paper wallets to store their cryptocurrencies securely. Hardware and software wallets relying on traditional cryptographic algorithms like RSA or ECDSA are vulnerable to quantum attacks. The quantum threat and impact on IDPs are similar to traditional authentication systems, as the security relies on the underlying cryptographic algorithms. OIDC relies on the security of the underlying cryptographic algorithms. If those algorithms are not quantum-resistant, there could be potential security risks in the future. Quantum threats extend beyond key management vulnerabilities to the broader blockchain network, jeopardizing both cryptographic security and the integrity of communication between nodes.

5.4. Network architecture

Quantum computers have the potential to break cryptographic algorithms used in blockchain networks, such as asymmetric encryption and digital signatures, which compromise the security and integrity of transactions and data stored on the blockchain, enabling malicious actors to forge transactions, manipulate data, or even take control of the blockchain network. Quantum computers can also compromise the security of communication channels between blockchain nodes. This includes the confidentiality and integrity of data transmitted between nodes, such as transaction details, blocks, and consensus-related information. Sophisticated quantum man-in-the-middle attacks can be carried out on blockchain node communication by intercepting and altering

the communication between nodes, and attackers can manipulate transactions, forge digital signatures, and tamper with the integrity of the network. This can lead to unauthorized funds transfers, data manipulation, and disruption of the blockchain network's operation [207]. Beyond compromising blockchain security and network communication, quantum threats also raise concerns about the reliability and resilience of smart contracts, which are already vulnerable to various security flaws at different levels of implementation.

5.5. Smart contract

Current vulnerabilities in smart contracts surveyed from 49 articles can be classified into three levels: language, virtual machine, and blockchain [12]. A survey conducted in [208] showed 15 security vulnerabilities in smart contracts before deployment exists. Vulnerability detection tools for smart contracts can be categorized into static analysis, dynamic analysis, and formal verification. The survey concludes that existing tools can only detect new attacks by requiring predefining the vulnerability first. Other work [209] suggested six categories of vulnerability detection tools for smart contracts: formal verification method, symbol execution method, fuzzy testing method, intermediate representation method, stain analysis method, and deep learning method. This work tested 27 detection tools and complemented previous work's finding that detection tools can only detect existing smart contract vulnerabilities.

6. Post-quantum cryptography (PQC)

The NIST has identified four candidate algorithms for standardization with one public-key encryption / key-establishment mechanism (KEM) and three digital signature algorithms. The four quantum-resistant algorithms rely on mathematical problems that both conventional and quantum computers should be computationally hard to solve. The general encryption algorithm secures the information exchanged over a public network and digital signatures used for identity authentication. SPHINCS+ uses hash functions, while CRYSTALS-Kyber, CRYSTALS-Dilithium, and SPHINCS+ are based on a family of math problems called structured lattices [101]. All four of the algorithms were created by experts collaborating from multiple countries and institutions [25]:

- The NIST has chosen the CRYSTALS-Kyber for general encryption. Its benefits include relatively compact encryption keys that two parties and its rapid operation can readily exchange. Kyber is an IND-CCA2-secure key encapsulation mechanism (KEM), the security of which relies on the difficulty of solving the learning-with-errors (LWE) problem over module lattices. It is believed to be resistant to attacks by both classical and quantum computers. Combining a symmetric encryption algorithm with a key encapsulation module, CRYSTALS-Kyber provides a hybrid encryption scheme [210].
- CRYSTALS-Dilithium, FALCON, and SPHINCS+ (pronounced "Sphincs plus") are the three algorithms designated by NIST for digital signatures. NIST recommends CRYSTALS-Dilithium as the primary algorithm and FALCON for applications requiring smaller signatures than Dilithium. Reviewers remarked on the high efficacy of the first two algorithms, and based on this, CRYSTALS-Dilithium is recommended as the primary algorithm. The third, SPHINCS+, is somewhat more extensive and slower than the other two, but it is a valuable fallback because it is based on a different mathematical approach than the other three NIST options.

CRYSTALS-Dilithium: Dilithium is predicated on the complexity of the Ring Learning with Errors (RLWE) problem, which is believed to resist attacks from classical and quantum computers. CRYSTALS-Dilithium offers robust security guarantees, such as impossibility and resistance to key recovery and substitution assaults. It offers a variety of

security levels and trade-offs between key size, signature size, and computational complexity, enabling deployment flexibility. The security concept dictates that an adversary with access to a signing oracle cannot produce a signature of a message whose signature one has not yet observed, nor can he produce a distinct signature of a message whose signature he has already observed [211].

FALCON: FALCON is founded on Gentry, Peikert, and Vakuntanathan's theoretical framework for lattice-based signature schemes. It is based on the problem of solving the Short Integer Solution (SIS) over NTRU lattices, for which no efficient solving algorithm is presently known in the general case, even when using quantum computers [212]. NTRU (Nth degree TRUncated polynomial ring) is an open-source public-key cryptosystem that uses lattice-based cryptography to encrypt and decrypt data introduced by Jeffrey Hoffstein, Jill Pipher, and Joseph H. Silverman in 1996.

SPHINCS+: SPHINCS+ is a post-quantum cryptographic algorithm for stateless hash-based digital signatures. It offers strong security against attacks from classical and quantum computers by utilizing a tree-based structure and employing hash functions as the underlying primitive. It provides provable security guarantees, including resistance against forgery and key recovery attacks. SPHINCS+ is designed to be lightweight and efficient, making it suitable for resource-constrained environments [213].

Some practical implementation and adoption approaches for quantum-safe solutions in blockchain can be categorized into lattice-based, code-based, multivariate, hash-based, isogeny-based, and FHE [38]. Table 10 provides a summary of these post-quantum cryptographic (PQC) methods, which are further detailed in the following sections.

6.1. Lattice-based

The key concept in lattice-based encryption is the difficulty of problems on lattices. A lattice consists of all possible vectors that can be generated using linear combinations of a given set of basis vectors. While lattices are often visualized in two or three dimensions for simplicity, practical cryptographic applications use high-dimensional spaces. Different sets of basis vectors can define the same lattice, as shown in Fig. 4 where blue and green vectors form the same lattice as the original red and black vectors. Lattice-based cryptography is a promising foundation for upcoming cryptographic methods since lattice-

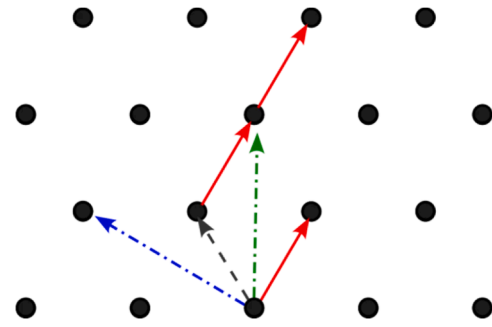


Fig. 4. Example of a lattice generated by two two-dimensional vectors, one depicted with solid red lines and the other with black dashed lines [217].

based mathematical problems are challenging to answer, even by quantum computers [214]. Lattice-based cryptography offers a wide range of PQC algorithms, including lattice-based signatures (e.g., NTRU, BLISS), encryption schemes (e.g., LWE, NTRU), and key exchange protocols (e.g., New Hope). Lattice-based cryptography is known for its strong security properties and resistance against quantum attacks [215]. It has gained significant attention as a potential PQC candidate for cryptographic applications, including blockchains. In [216], a lattice-based ring signcryption for consortium blockchain is constructed and shows the scheme to protect user privacy by reducing the connection between blockchain and user information and quantum-safe.

6.2. Code-based

A key component of code-based cryptography is the difficulty of decoding error-correcting codes. It provides several PQC schemes, including Niederreiter and McEliece encryption. With its lengthy research history, code-based cryptography offers a viable defense against quantum attacks. It might, however, perform slower and have larger key widths than other PQC substitutes [215]. Alahmadi et al. [218] presented a novel digital signature system based on error-correcting codes for blockchain technology. Signatures are validated using the limited distance decoding technique. The sizes of the signed message, the message, and the transmission rate have all been computed. The suggested strategy is more beneficial in the industry because of its short signature length. Another advantage of this approach is its high transmission rate. This reduces the amount of money and time needed to verify the signature.

6.3. Multivariate polynomial-based

The foundation of multivariate cryptography is the difficulty of solving multivariate polynomial equations over a finite field. The multivariable polynomial problem is suitable for PQC because it is non-deterministic polynomial-time (NP)-complete [219]. It is common knowledge that this method works very well for cryptanalysis creation. The digital signature creation produces the smallest digital signature produced by the PQC algorithms. Schemes like Rainbow, TTS, QUARTS, and QUAD are among them [215]. A novel proof of work (PoW) for blockchain was put out by Ding [220] and is based on random multivariate quadratic equations. It is based on how difficult it is to solve a set of random quadratic equations over a finite field. The suggested Proof of Work (PoW) has already been incorporated into the recently launched cryptocurrency ABC and is said to offer benefits over the conventional PoW.

6.4. Hash-based

One such possibility for post-quantum cryptography is the hash-based signature. Digital signature methods that use hash functions as

Table 10
Summary of cryptographic methods.

Method	Principle	Strengths	Weaknesses
Lattice-based	Hardness of lattice problems	High security, quantum-resistant, widely studied	Larger key sizes, computationally intensive
Code-based	Difficulty of decoding error-correcting codes	Long research history, strong post-quantum security	Slow performance, large key sizes
Multivariate Polynomial-based	Solving multivariate polynomial equations over a finite field	Efficient cryptanalysis creation, small signature sizes	Complex implementation, susceptible to attacks
Hash-based	Security of hash functions	Mature, strong security, well-established	Inefficient for frequent key updates
Isogeny-based	Mathematical structure of elliptic curves and isogenies	Potential replacement for Diffie-Hellman encryption	Efficiency bottleneck, slow adoption
Fully Homomorphic Encryption (FHE)	Performing computations directly on encrypted data	Allows computation on encrypted data without decryption	High computation overhead, slow performance

their main building component are known as hash-based signature schemes. Hash-based encryption is based on the security of hash functions and a few renowned hash function-using algorithms, such as MD5 and SHA [215]. Like Lamport and Merkle signatures, Hash-based signature schemes are among the most mature and well-studied PQC alternatives. They are based on the one-wayness of hash functions and offer strong security against quantum attacks. However, they are less efficient for scenarios that require frequent key updates. The quantum hash function based on controlled alternate quantum walks proposed in [221] yields exceptional collision resistance, message sensitivity, diffusion, and confusion performance. Recent hash-based signature schemes include XMSS (eXtended Merkle Signature Scheme), Leighton-Micali signature (LMS), SPHINCS, and BPQS. When a key is reused for a reasonable number of signatures, BPQS performs better than current hash-based algorithms (ECDSA SECP256K1, EdDSA Ed25519, RSA3072, and SPHINCS-256). Additionally, it provides a fallback mechanism that enables a practically infinite number of signatures if necessary [222].

6.5. Isogeny-based

Isogeny-based cryptography utilizes the mathematical structure of elliptic curves and their isogenies. Notable schemes in this category include SIDH (Supersingular Isogeny Diffie-Hellman), IbdS (Isogeny-Based Digital Signature), SDVS (Strong Designated Verifier Signature), undeniable signatures, SIOT (Supersingular Isogeny Oblivious Transfer), and SIKE (Supersingular Isogeny Key Encapsulation). The characteristics of SIDH suggest that it could be a suitable replacement for the widely used Diffie-Hellman and elliptic curve Diffie-Hellman key exchange. Similarities between SIDH and DH led to the quick development of isogeny-based cryptosystems employing classical cryptography-inspired architectures. Since SIDH and DH are comparable, they can be combined to provide public key encryption that resembles ElGamal public key encryption [223]. A novel approach to isogeny computation using Kummer lines proposed in [224] increases the efficiency of isogeny-based cryptography, which exposes its cryptographic applications. Wider adoptions of isogeny-based cryptography require work on handling the efficiency bottleneck.

6.6. Fully homomorphic encryption

Considering server-side privacy approaches, the homomorphic encryption (HE) systems allow computations directly on ciphertexts by reflecting the appropriate actions on the plaintexts. The idea of homomorphic encryption is expanded upon with Fully Homomorphic Encryption (FHE), which permits arbitrary calculations on encrypted data. This means that any sequence of operations, including addition, multiplication, and more complex functions, can be performed on encrypted data and still get an encrypted result corresponding to those operations' outcomes on the unencrypted data [225,226]. Ideal and leveled FHE are two subcategories of Fully Homomorphic Encryption (FHE) algorithms. In contrast to leveled FHE schemes, which can only support a finite amount of calculations based on a specified polynomial-size circuit, ideal FHE schemes can allow calculations for any circuit, including circuits with an infinite number of operations on crypto texts [227]. A work in [228] proposed fully homomorphic encryption as a privacy-protection solution that encrypts data on chains to make it "available and invisible" for on-chain privacy data.

7. Quantum-safe solutions

The adoption of quantum-safe solutions in blockchain faces critical challenges, including secure random number generation, efficient key distribution, quantum-resistant consensus mechanisms, authentication protocols, and robust communication frameworks. Quantum random number generators (QRNGs) enhance cryptographic security by leveraging quantum mechanics for true randomness, mitigating

vulnerabilities in traditional RNGs. Quantum key distribution (QKD) strengthens blockchain security by enabling tamper-resistant key exchange, ensuring protection against quantum-enabled attacks. Quantum-resistant consensus mechanisms, such as quantum zero-knowledge proofs and quantum-enhanced Proof of Stake (QDPOS), are being developed to withstand quantum adversaries while optimizing efficiency, ensuring both decentralization and security. Quantum authentication protocols, including quantum identity authentication and quantum deniable authentication, provide secure identity verification in decentralized networks, reducing risks of unauthorized access and fraud. Additionally, quantum internet protocols, such as quantum-secure transmission (qTCP) and quantum repeater networks, are essential for securing blockchain communication in a quantum-driven environment.

7.1. Quantum random number generators (QRNGs)

Random number generators (RNGs) are important for blockchain technology for consensus mechanisms, random selection of nodes for various tasks, randomness in smart contracts, decentralization degree of blockchain applications, security, and privacy. Blockchains often require a random selection of nodes for tasks such as shuffling data, distributing rewards, and participating in leader elections. Randomness in smart contracts is required for games, for example, where RNG ensures that the outcomes are unpredictable and unbiased, maintaining the integrity and fairness of the application. Various cryptographic operations that constitute blockchain systems require RNGs for generating encryption keys, digital signatures, and unique identifiers. Related works on the QRNG are listed and summarized in Table 11.

7.2. Quantum key distribution (QKD) protocols

Blockchain security depends on key distribution mechanisms to maintain the integrity of digital assets and transactions. Decentralized networks' identification, access control, and data integrity are all based on cryptographic keys. Strong key distribution procedures guard the security architecture, avoiding fraud, data breaches, and declining user and participant trust. By using the ideas of quantum mechanics, two parties, usually known as Alice (the sender) and Bob (the receiver), can establish a shared secret key through a secure communication technique called quantum key distribution (QKD). Because QKD is based on the fundamental laws of quantum mechanics—specifically, the no-cloning theorem and the principles of quantum superposition—it is unique. Because of these characteristics, any attempt to intercept the communication will cause the transmitted quantum states to fluctuate, allowing the people involved to detect it. Various QKD approaches have been developed in recent years, i.e., quantum bit commitment, quantum secret sharing, quantum coin flipping, quantum fingerprinting, quantum digital signatures, blind quantum computing, and position-based quantum cryptography [48]. Related works on the QKD are listed and summarized in Table 12.

7.3. Quantum-resistant consensus mechanisms

Consensus mechanisms are the foundation of blockchain technology, enabling a decentralized network to agree on the ledger's state without a central authority. They prevent malicious actors from manipulating the blockchain and maintain data consistency. Without robust consensus, blockchain systems are vulnerable to double-spending, fraud, and data inconsistencies, making them unreliable and untrustworthy for various applications. Quantum-resistant consensus mechanisms are blockchain consensus protocols designed to remain secure even in the presence of powerful quantum computers. Related works on the quantum-resistant consensus mechanism are listed and summarized in Table 13.

Table 11
Related works on QRNGs.

Year	Refs.	Title	Outlook
2023	[39]	A quantum circuit to generate random numbers within a specific interval	The circuit is fault-tolerant, optimized for qubits, T-count, and T-depth, and customizable for user-specified intervals. Its effectiveness is tested, with a 90 % or 95 % success probability. A reversible comparator is also presented, valid for other quantum circuits and algorithms, and the best available in the literature in terms of qubits, T-count, and T-depth.
2023	[40]	Optical quantum random number generators: a comparative study	A discrete quantum random number generator can be created using a homodyne detector, with random number generation rates increased by ASE (amplified spontaneous emission) from and EDFA (Erbium-doped fiber amplifier) and better entropy values and random generation rates achieved using two lasers.
2022	[41]	Quantum true random number generation on IBM's cloud platform	This work designs and tests a random number produced on an IBM quantum platform, assuming identical gates. The system produces over 20 million random bits on the cloud, passing all NIST 800-90B and 22 test suite substances without postprocessing. The output bit strings have outstanding randomness quality. The system is practical and easier to implement than physical device-independent RNG. It can be used in end-to-end secure communications and offered as an Entropy as a Service (EaaS) for cloud service providers.
2021	[42]	Quantum generators of random numbers	This paper highlights the importance of unconditional randomness in quantum random number generators, contrasting with pseudorandom classical generators. It identifies sources of entropy in quantum measurements and transitions using the Fermi golden rule. Two prototypes are proposed, removing bias and passing randomness tests in NIST and Dieharder batteries. A new algorithm for non-destructive public randomness testing is proposed, overcoming local computational resource limitations and reducing complexity.
2020	[43]	Quantum random number generation	Quantum random number generators use quantum mechanics principles to generate true random numbers. Different QRNG products use shot noise, MOSFET transistors, or photons as entropy sources. Quantis QRNG uses single-photon splitting, while the Australian National University uses quantum vacuum fluctuations. PicoQuant and qutool QRNGs use the photon arrival time, while QuintessenceLabs QRNG offers the fastest speed of 1 GBps using quantum vacuum fluctuations. The maximum number of QRNGs available is based on quantum optics.

Table 12
Related works on QKDs.

Year	Refs.	Title	Outlook
2023	[44]	End to end secure e-voting using blockchain & quantum key distribution	This paper explores the shortcomings of blockchain technology brought forth by quantum computing and makes recommendations for enhancements. It suggests a blockchain-secured electronic voting system that is constructed using Python on the Linux operating system with quantum key distribution. The system seeks to meet the needs and specifications of the electronic voting procedure.
2021	[45]	Quantum key distribution with two-way authentication	This paper presents a quantum key distribution protocol with two-way authentication, transforming symmetric and asymmetric key authentication systems. It aims to achieve quantum key distribution with quantum authentication, demonstrating the potential for wider applications.
2020	[46]	Optimization and implementation of efficient and universal quantum key distribution	This paper optimizes EUQKD against bit errors, improving the QKD platform's efficiency. The protocol is implemented on a BB84 QKD platform, reducing error bit impact and generating set of bits (Cbts) with exceptional statistical properties, thereby preventing protocol failure.
2019	[47]	Blockchain based secure crypto-currency system with quantum key distribution protocol	This research supports blockchain development by discussing quantum mechanics, cryptography, and the six-state Quantum key distribution protocol. Simulations on a six-state mathematical model support this, paving the way for future implementation of key generation protocols.
2016	[48]	Practical challenges in quantum key distribution	This review discusses challenges in practical QKD, including extending security proofs, general attacks, photonic chips, and global-scale networks. Advances in QKD will impact everyday interactions, quantum communication protocols, and the future quantum internet.
2015	[49]	Quantum key distribution: simulation and characterizations	As an alternative to established key distribution algorithms, this paper introduces QKD, a novel cryptographic key generation technology. It examines configurations such as initial qubit length, noise level, and attack influences, and uses two modes to perform the QKD-BB84 protocol. QKD is compatible with a wide range of security apps and contemporary encryption techniques.

7.4. Quantum authentication protocols

Authentication protocols are crucial for blockchain due to their role in verifying the identity of participants within decentralized networks. They ensure that only authorized users can access and engage with the blockchain, maintaining data integrity and security. In a trustless

Table 13
Related works on quantum-resistant consensus mechanisms.

Year	Refs.	Title	Outlook
2022	[50]	Blockchain consensus mechanism based on quantum zero-knowledge proof	In contrast to conventional blockchain consensus mechanisms, a quantum blockchain consensus mechanism can withstand a 51 % attack and offers lower energy consumption, shorter latency, greater throughput, and is based on the unpredictability and irreversibility of quantum measurement and quantum zero-knowledge proof.
2022	[51]	A blockchain consensus protocol based on quantum attack algorithm	The paper introduces a novel consensus protocol that uses quantum measurement's stochasticity, irreversibility, and uncertainty to enhance authentication communication. This protocol ensures ciphertext security, reducing the need for mathematical proofs and computations. Compared to traditional encryption algorithms, it requires minimal computing resources, consumes minimal energy, and achieves high throughput. Experimental results show that it improves blockchain resistance to quantum attacks.
2022	[52]	Quantum consensus dynamics by entangling Maxwell demon	The work presents a Maxwell demon that creates and maintains a many-body entanglement in a functional material. It makes sense of quantum dynamics by applying a quantum version of the noisy voter model. Additionally, the study developed the second law of thermodynamics, which states that entangling devils can only be defeated in situations where information gain exceeds absolute irreversibility. Any quantum system can use the protocol.
2022	[53]	On the robustness of quantum algorithms for blockchain consensus	The research assesses entanglement-free quantum-based algorithms and quantum consensus for realistic blockchain-enhanced distributed networks. Comparable decentralization, reduced scalability, and improved fault tolerance characterize quantum consensus. Although they are not as scalable, entanglement-based algorithms provide higher security. Only modular computing units and small-scale distributed sensor networks are viable for current quantum protocols.
2022	[54]	Efficient quantum blockchain with a consensus mechanism QDPoS	Quantum blockchain offers an alternative to classical blockchain to resist malicious attacks from future quantum computers. However, current schemes lack a suitable consensus mechanism, leading to low efficiency. Quantum delegated proof of stake (QDPoS) is proposed to provide fast decentralization and efficient chain formation. This design enhances adaptability to the quantum era's challenges.
2019	[55]	Quantum consensus	Quantum distributed ledgers can improve intra-network communication and interoperability models, offering

Table 13 (continued)

Year	Refs.	Title	Outlook
			improved speed, scalability, and security. With rapid technology growth, quantum computers could become commercialized, allowing corporations to use quantum algorithms for competitive advantages. However, zero-knowledge proofs are not quantum secure and require large cryptographic proof size and computing resources, making existing solutions unfavorable for distributed ledgers reliant on communication networks.

environment where users transact pseudonymously, authentication protocols provide the means to establish trust and accountability, preventing unauthorized access, fraud, and malicious activities. Robust authentication also enables secure management of cryptographic keys, which is fundamental to blockchain's security and data protection. Authentication protocols are a cornerstone of blockchain's functionality, fostering trust, enabling secure transactions, and safeguarding the integrity of distributed ledger systems across various applications. Related works on the quantum-resistant consensus mechanism are listed and summarized in [Table 14](#).

7.5. Quantum internet protocols

Internet protocols are essential for blockchain as they allow blockchain nodes to communicate and synchronize across the global network. These protocols, like TCP/IP and HTTP, enable the secure transmission of blockchain data, ensuring that transactions and blocks are propagated accurately and reliably. Internet protocols also facilitate the connection of nodes, allowing them to reach consensus, validate transactions, and maintain the integrity of the blockchain ledger in a decentralized manner. These protocols are necessary for blockchain networks to be isolated and able to function as distributed and immutable ledgers, hindering their capacity to provide secure and transparent transaction records across a broad range of applications and industries. Related works on the quantum internet protocols are listed and summarized in [Table 15](#).

8. Quantum-safe blockchain

8.1. Quantum-safe blockchain implementations

Quantum-safe blockchain platforms are engineered to address this emerging threat, ensuring blockchain technology remains secure and reliable in a post-quantum era. Leveraging PQC algorithms and other post-quantum solutions, these platforms provide robust defenses against quantum attacks while preserving the core principles of blockchain's transparency, decentralization, and immutability. Notable projects and initiatives in the field are QAN (Quantum-resistant, Anonymous, Non-forking) and QRL (Quantum Resistant Ledger) as described in [Table 16](#).

8.2. Research and development in quantum-safe blockchain

Researchers are actively designing and testing quantum-resistant algorithms and protocols, striving to integrate them into existing blockchain networks or create entirely new quantum-safe blockchains. Other relevant works on the quantum-safe blockchain are listed and compared in [Table 17](#).

Table 14
Related works on quantum authentication protocols.

Year	Refs.	Title	Outlook
2023	[56]	Quantum identity authentication based on the extension of quantum rotation	The foundation for quantum rotation protocols is covered in this paper, which addresses the geometric characteristics of quantum rotations and their dimensionality. A proposed protocol for bit-oriented QIA integrates quantum rotation superposition with the QPKC architecture. Every bit of the protocol is very secure and resistant to numerous assaults; the security of each bit is dependent on the length of the authentication code. The protocol is appropriate for quantum computing networks since it is effective, easily extendable, and does not require additional quantum resources.
2023	[57]	A novel quantum identity authentication protocol without entanglement and preserving pre-shared key information	In this research, a novel quantum authentication mechanism utilizing single photons for quantum secure direct communication is proposed. With safe message encryption, this protocol may confirm user identity without disclosing the pre-shared key. Eve's presence is verified via decoy states and random authentication keys. The protocol does not require quantum memory and is secure against impersonation, intercept measure-resend, and entangle-measure attacks.
2021	[58]	Experimental authentication of quantum key distribution with post-quantum cryptography	In the age of quantum computing, safe key exchange is made possible through quantum key distribution (QKD). However, it needs $C_n^2 = n(n-1)/2$ pairs of symmetric keys for authentication using traditional channels. Quantum-resistant security combined with post-quantum cryptography (PQC) can offer effective and safe authentication for QKD. Benefits for novice users are demonstrated by the experimental validation of PQC's viability, effectiveness, and stability in QKD authentication. Applications for quantum-safe communication can be expanded by combining QKD and PQC authentication. This study provides a two-mode squeezed quantum state-based non-interactive quantum deniable authentication system. An initial secret key that has been translated into two-mode squeezed states is shared by Alice and Bob. Identity authentication is made possible by balanced homodyne measurement findings, which allow both parties to deduce hidden information. The correlation qualities of the
2020	[59]	A quantum deniable authentication protocol based on two-mode squeezed quantum states	

Table 14 (continued)

Year	Refs.	Title	Outlook
			states and random numbers ensure the security of the protocol.

8.3. Industry adoption and standardization efforts

Quantum-safe blockchain platforms not only secure transactions against quantum attacks but also set the stage for the next evolution of digital infrastructure. Their development is spurring significant research and development efforts aimed at integrating post quantum cryptographic algorithms seamlessly into existing blockchain protocols. This evolution requires rethinking current consensus mechanisms and communication protocols, as well as developing new frameworks that can maintain high performance and scalability while ensuring robust quantum security. Researchers are exploring novel techniques to balance the trade-offs between enhanced security and system efficiency, addressing issues such as increased computational overhead and potential bottlenecks in transaction throughput.

In parallel, industry adoption of quantum-safe blockchains is gathering momentum. Pilot projects and experimental deployments are being conducted across various sectors—from finance and healthcare to supply chain management and IoT ecosystems. These initiatives not only validate the practical feasibility of quantum-resistant techniques but also highlight the economic and regulatory benefits of transitioning to quantum-safe systems. As stakeholders become increasingly aware of the potential risks posed by quantum computing, there is a growing impetus to update legacy systems and ensure long-term data integrity and privacy.

Furthermore, collaborative efforts between academia, industry, and government agencies are critical in advancing the standardization of quantum-safe cryptographic protocols. International bodies, such as the National Institute of Standards and Technology (NIST), are actively working to vet and standardize algorithms that can be deployed across diverse blockchain platforms. These standardization efforts are expected to provide a unified framework for implementing quantum-resistant solutions, thereby reducing fragmentation and ensuring interoperability across different systems.

8.4. Future directions in quantum-safe blockchain

As quantum computing continues to advance, the long-term security of blockchain networks will depend on the successful integration of quantum-safe cryptographic techniques. Future research and development efforts in quantum-safe blockchain will focus on several key areas, including the implementation of quantum key distribution (QKD), advanced post-quantum digital signatures, hybrid cryptographic models, quantum-safe consensus mechanisms, and scalable blockchain architectures. These advancements will ensure that blockchain networks can resist quantum threats while maintaining efficiency, decentralization, and scalability.

One of the most promising quantum-safe techniques is Quantum Key Distribution (QKD), which enables the secure exchange of cryptographic keys between blockchain nodes using the principles of quantum mechanics. Unlike classical key exchange protocols that rely on computational hardness assumptions, QKD ensures unconditional security based on the no-cloning theorem and quantum superposition. Future research will focus on integrating QKD into blockchain networks to provide secure communication channels, preventing quantum-enabled eavesdropping and man-in-the-middle attacks. Additionally, efforts are being made to develop satellite-based QKD networks [231], which will allow blockchain nodes across the globe to exchange cryptographic keys securely, further enhancing the resilience of decentralized networks.

Table 15
Related works on quantum internet protocols.

Year	Refs.	Title	Outlook
2022	[60]	Quantum Internet protocol stack: A comprehensive survey	Over the past 50 years, the traditional Internet has undergone substantial evolution, with multiple layers offering communication services. Nevertheless, the Quantum Internet is not easily adaptable because of quantum mechanics. To fully utilize quantum entanglement and information, the architecture of the Quantum Internet necessitates a paradigm change. This survey focuses on the unresolved issues and necessary work toward a functional and comprehensive quantum Internet protocol stack.
2021	[61]	Protocols for packet quantum network intercommunication	The article suggests qTCPS for dependable quantum packet communication and explores how packet quantum networks communicate for quantum repeater networks. The quantity of shares needed for quantum messages or secret-sharing recovery procedures is the primary expense of the retransmission protocol. Examining congestion avoidance and control strategies for qTCPS is the next stage.
2021	[62]	Quantum Internet—Applications, functionalities, enabling technologies, challenges, and research directions	Researchers are investigating quantum networks for security, privacy, and capabilities including quantum computation, communication, and metrology because traditional Internet technology, which is based on mathematics and Claude Shannon's theory of information, is susceptible to eavesdroppers. In order to establish a Quantum Internet (QI), distant nodes must communicate over quantum channels that are secured by quantum cryptography algorithms. With their teleportation, entanglement, and superposition capabilities, qubits are advantageous over conventional networks. For the physical manifestation of QI, research is being done on satellite-based quantum communication.
2018	[63]	Quantum internet: A vision for the road ahead	It takes interdisciplinary efforts in computer science, engineering, and physics to build and scale quantum networks. The stages that have been suggested will help with communication

Table 15 (continued)

Year	Refs.	Title	Outlook
			and offer instructions for hardware implementation, software development, and protocol design. In the coming years, multi-node quantum networks will emerge, offering an exciting chance to test concepts and features.

Table 16
Comparison between QAN and QRL.

QAN (Quantum-resistant, Anonymous, Non-forking)	QRL (Quantum Resistant Ledger)
The QAN blockchain platform is a quantum-resistant Layer 1 hybrid blockchain platform that allows developers to code in any language. The QANplatform public network can process up to 1600 transactions per second, and 97,000 transactions per second in a private network. It claims to share the same decentralization level as Bitcoin and Ethereum and is compatible with Ethereum EVM. The platform's utility token is called QANX to be used to pay for contract deployment, executing methods, transaction fees, validation tickets (to participate in the validation process), full-node license (to be a full-node provider), and smart contract rewards [229].	The blockchain platform QRL was created with resistance against quantum computing in mind. It uses XMSS, a post-quantum signature technique, to protect digital signatures from assaults by both classical and quantum computers. Cryptographic hazards, both known and undiscovered, are shielded from the QRL digital asset. The digital assets of QRL are secured via public and private post-quantum security, which is built on top of the core protocol. Post-quantum communication can be achieved by combining on-chain lattice key storage with the powerful short-lived messaging layer [230].

9. Discussion

The transition to quantum-safe technologies for blockchain is crucial due to the advancing field of quantum computing and its potential to disrupt existing cryptographic systems. Blockchain technology is heavily dependent on existing traditional cryptography and security protocols, which have been presented throughout this paper to be highly vulnerable to quantum computer attacks. As quantum computers become available and affordable to the general public, they could undermine the security of existing blockchain platforms and allow attackers to disrupt the entire blockchain system. The trustless property of decentralized ledger technology, such as blockchain, requires a security guarantee for broader blockchain applications to critical applications like finance, healthcare, and governance. Regulatory bodies are likely to impose stricter security standards as blockchain technology matures. Hence, adhering to post-quantum security measures can help blockchain projects comply with evolving regulations and demonstrate their commitment to protecting user data. Implementing these changes as soon as possible ensures forward compatibility with quantum-resistant algorithms and other solutions to minimize disruptions when quantum computers become a reality.

Transitioning into a quantum-safe blockchain faces numerous challenges and open issues, i.e., the need for standardization, integration complexity, and performance concerns. There still needs to be standardized post-quantum cryptographic algorithms for blockchain applications, making it challenging for developers to choose the most suitable ones. Integrating PQC, QRNG, QKD, quantum-resistant consensus mechanisms, quantum authentication protocols, and quantum internet protocols into existing blockchain platforms is complex and resource-intensive. Some solutions are computationally intensive, if not requiring specific quantum computers or devices to achieve, which can affect the performance and scalability of blockchain applications. A

Table 17
Comparison of quantum-safe blockchain.

Year	Refs.	Title	Outlook
2023	[38]	Quantum-resistance in blockchain networks	The study describes a post-quantum blockchain network that included a Falcon-512 post-quantum signature in transactions, attained quantum resistance in communications, and used a quantum source of entropy. Solidity verification code, EVM assembly language operation code, and native smart contract are the three methods for verification that the team suggests. In order to use new post-quantum signature cryptographic algorithms, the article advises altering Ethereum protocols, which could increase security. Another blockchain network can use the same framework; however, the technology used to implement it will differ. The vulnerability of blockchain networks to hacking by quantum computers is highlighted in the report as a reason for their protection.
2023	[31]	Post quantum blockchain architecture for internet of things over NTRU lattice	The paper proposes an NTRU lattice-based blockchain system for IoT, which is secure against quantum computing attacks and efficient in resource-constrained environments. It presents a seed key generation algorithm and two improvement schemes. However, open problems include compressing block content without affecting security, designing lattice signatures efficiently, and introducing privacy protection technologies to improve the system's privacy.
2022	[32]	Post-quantum hybrid encryption scheme for blockchain application	The study demonstrates that a hybrid encryption scheme using CRYSTALS-Kyber and AES-256 is a feasible solution for blockchain applications to withstand quantum attacks. It was tested on the Oricon blockchain application, showing minimal modifications needed. The CRYSTALS-Kyber 768 key size was found to be the most optimal, with stable encryption and decryption time, peak successful transactions, and low latency. However, implementing this key size decreased throughput and performance.
2022	[33]	Quantum blockchain based on asymmetric quantum encryption and a stake vote consensus algorithm	The proposed quantum blockchain algorithm improves fairness, efficiency, and security by generating blocks and signing transaction information using a quantum one-way function. It addresses the QSCDff problem and penalizes malicious behaviors. The algorithm offers security in semi-honest and malicious adversary models. It also

Table 17 (continued)

Year	Refs.	Title	Outlook
			provides a safe platform compared to other quantum blockchain algorithms. Future research could develop quantum signatures without a trusted node and quantum privacy computing to enhance security.
2022	[34]	Decentralization using quantum blockchain: A theoretical analysis	This article explores the potential of quantum blockchains for decentralized quantum identity, a concept designed by integrating existing quantum approaches like PKIs and blockchains. Despite technical gaps, the article believes a decentralized perspective on quantum applications is significant and likely.
2022	[35]	A quantum blockchain-enabled framework for secure private electronic medical records in Internet of Medical Things	The research proposes a quantum electronic medical record system for healthcare, addressing the vulnerability of current blockchain systems to quantum computer attacks. The system uses entangled states, quantum authentication, and a quantum circuit diagram for hash value generation, ensuring security and privacy in the Internet of Medical Things.
2021	[36]	Quantum blockchain system	Blockchain technology like Bitcoin and Ethereum has applications in digital currency and IoT. However, security depends on computational complexity. With quantum computers, security is threatened. A quantum blockchain system with quantum secure communication is proposed, ensuring unconditional security in classical cryptographic theory.
2019	[37]	Quantum blockchain: A decentralized, encrypted and distributed database based on quantum mechanics	Quantum blockchain, like classical blockchain, offers decentralization and security features. Its main advantages include safety and efficiency. Quantum secure direct communication (QSDC) and quantum key distribution (QKD) ensure communication security between nodes. Digital signatures, like RSA, can be vulnerable to quantum computer attacks. Quantum digital signature schemes can solve this issue. The blockchain also has a fast transaction processing speed, with the POW based on Analog Hamiltonian Optimizations promoting cryptocurrency development. However, access to universal quantum computers may give accessors an unfair advantage in mining awards.

hybrid approach of adding quantum-safe solutions to existing blockchain security is achievable until access to quantum computers is more common.

Using available quantum-safe blockchain platforms instead of

transitioning from traditional blockchains has advantages and disadvantages. One of the most significant advantages of using quantum-safe blockchains is that they provide immediate security against quantum attacks. These blockchains are designed with post-quantum cryptographic algorithms from the start, ensuring that the network is resilient to quantum computing threats right from the beginning. Transitioning from traditional blockchains to quantum-safe ones can be complex and time-consuming. Organizations can avoid the hassle and potential disruptions associated with migration by using an existing quantum-safe blockchain. Users and stakeholders can have greater trust in a blockchain that is already quantum-safe. This trust can be especially crucial in applications where security and data protection are paramount, such as in the financial industry or critical infrastructure. Established quantum-safe blockchains often have a mature ecosystem with developers, users, and support systems in place. This can provide a smoother experience for organizations adopting quantum-safe blockchain technology.

The primary drawback of relying on existing quantum-safe blockchains is the limited number of options available. Current major blockchain platforms and technologies in terms of market capitalization are based on non-quantum-safe security. This limited choice could lead to a need for more flexibility in selecting a blockchain platform that aligns perfectly with an organization's needs. In addition, shifting to a quantum-safe blockchain from a traditional one requires significant adjustments to their existing systems and processes, which could lead to additional costs and complexities. The field of quantum-safe cryptography and related solutions for blockchain itself is still evolving, and new cryptographic algorithms are being developed. Depending on the chosen blockchain platform, there could be a need to update or migrate to newer quantum-resistant algorithms in the future, which might introduce some degree of uncertainty.

10. Conclusion

This paper surveys the blockchain security infrastructure comprising cryptographic algorithms, consensus protocols, key and identity management, network architecture, and smart contract deployment and execution. These elements constitute the security of a blockchain system, one of the three most important keys for blockchain applications, along with decentralization and scalability. The advancement of quantum computing has upscaled existing threats and attacks toward any blockchain security infrastructure, exposing new vulnerabilities and escalating risks of quantum attack exploitation in the near future. Various quantum-safe security solutions are discussed and evaluated in relevance to blockchain security infrastructure: post-quantum cryptography, quantum random number generator, quantum key distribution, quantum-resistant consensus mechanisms, quantum authentication protocols, and quantum internet protocols. Recent works in those fields revealed significant progress made on quantum computing to be more applicable and usable in everyday applications. The availability of these works enables a hybrid approach to reinforce existing blockchain security infrastructure in the face of the quantum computing era. Recent developments on quantum-safe blockchains provide an alternative for future development of blockchain applications to embrace the quantum computing era and avoid the challenges and difficulties of transitioning from existing traditional blockchains to quantum-safe blockchains.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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